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2 Personalized AI assistants could either promote or hinder the
3 emergence of collective intelligence [SWP: *The potential*
4 *impact of personalized AI assistants on collective*
5 *intelligence?*]

6
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11 Abstract

12 Recent advances in artificial intelligence (AI) have enabled widespread use of per-
13 sonal AI assistants for various cognitive tasks, ranging from simple everyday tasks
14 to scientific discovery and creative writing. While AI assistance can improve the
15 performance of each individual, it can weaken their subsequent performance with-
16 out AI and further limit the diversity of interests and opinions. These risks raise
17 the question of whether human collective intelligence can persist as our society be-
18 comes increasingly reliant on AI assistants. To address this question, we develop an
19 evolutionary model of collective intelligence, where each individual participate in a
20 prediction task, consult AI, and imitate each other’s prediction strategy. We find
21 that “Chatbot” AI, which depends on users’ interests, can prevent the emergence
22 of collective intelligence by either limiting diversity or propagating systematic bias
23 through overreliance. Directly incentivizing or penalizing AI use generally worsens
24 these outcomes. Instead, a balanced incentive structure that preserve minority ex-
25 pertise and reward contributions to collective outcome can promote the emergence of
26 collective intelligence. However, the inherent ability of the population’s to make pre-
27 diction in the absence of AI remains suboptimal even under this balanced incentive
28 structure, raising concerns for continued AI reliance. [SWP: *conclusion?*]

29 Introduction

30 Collective intelligence (CI) describes a phenomenon in which a group’s ability to
31 solve problems, make decisions, or adapt to changing environments exceeds each
32 member’s ability (Galton, 1907; Woolley et al., 2010; Yaniv and Milyavsky, 2007;
33 Hommes et al., 2005; Lorge et al., 1958). This phenomenon, also referred to as
34 wisdom of crowds, has thus served as a central mechanism underlying the success of
35 diverse organizations, ranging from animal groups (Dreyer et al., 2025; Simons, 2004;
36 Berdahl et al., 2013) and human societies (Woolley et al., 2010; Riedl et al., 2021) to
37 machine-learning systems (Breiman, 1996, 2001). Understanding the conditions that
38 promote and maintain CI therefore remains a fundamental question across social
39 sciences (Mann and Helbing, 2017), evolutionary biology (Wang et al., 2025), and
40 computer science (Werfel, 2026). Addressing this question has become particularly
41 timely as artificial intelligence (AI) has begun to reshape how individuals acquire
42 information, make decisions, and contribute to collective outcomes (Cui and Yasseri,
43 2024).

44 The recent emergence of generative AIs and large language models (LLMs) has
45 changed the way individuals engage in various cognitive tasks, ranging from simple
46 everyday tasks to more complex tasks (Noy and Zhang, 2023; Brynjolfsson et al.,
47 2025), including scientific discovery (Lu et al., 2026) and creative writing (Doshi
48 and Hauser, 2024). In particular, while the use of generative AI models, such as
49 ChatGPT, has been rapidly increasing (Bick et al., 2026; Sajadieh et al., 2026), the
50 potential long-term impact of personal AI assistants on human society, especially the
51 CI, remains uncertain. Some argue that AI can promote CI by enhancing individual
52 expertise (Vaccaro et al., 2024), and opinion aggregation (Burton et al., 2024; Tessler
53 et al., 2024). Conversely, empirical evidence indicate that the use of AIs can limit the
54 diversity of opinions (Fügener et al., 2021; Doshi and Hauser, 2024; Lee and Chung,
55 2024; Meincke et al., 2025), a key requirement for the emergence of CI (Hong and
56 Page, 2004; Lorenz et al., 2011; Aplin et al., 2014; Becker et al., 2017).

57 Our ability to predict the long-term impact of personal AI assistants on human
58 CI is further hampered by the apparent discrepancy in the incentives that shape AI
59 uses. On one hand, many industrial sectors are pushing towards increased AI usages,
60 where the number of “tokens” used is often perceived as a proxy for productivity
61 (Bousquette, 2026; Keary, 2026). On the other hand, many academic institutions
62 appear to discourage, and even penalize, AI uses, reflecting concerns about plagia-
63 rism, misinformation, and limited independence (Thorp, 2023; Chawla, 2026). Such
64 discrepancy raises a question about whether directly incentivizing or penalizing AI
65 usage will benefit or harm human CI. These observations further highlight the im-
66 portance of indentifying a robust incentive structure that can promote CI in the
67 presence of personal AI assistants. Such incentive structure should seek to retain
68 the benefits of AIs and prevent potential risk, such as preserving the diversity of
69 individual opinions.

70 To address these questions, we extend previous approaches for modeling CI, where

each individual participates in a prediction task from partial observations of underlying causal factors (Mann and Helbing, 2017; Wang et al., 2025; Hong et al., 2012). In particular, we allow each individual to independently consult with personal AI assistants and revise their predictions accordingly. The resulting model allows us to investigate the robustness of human CIs under different assumptions about the AI model’s ability to make predictions as well as incentive structures that are associated with the prediction task. In doing so, we assess the impact of directly incentivizing or penalizing AI uses on CI and further suggest a balanced incentive structure that can maintain CI in the presence of AI.

Mathematical model

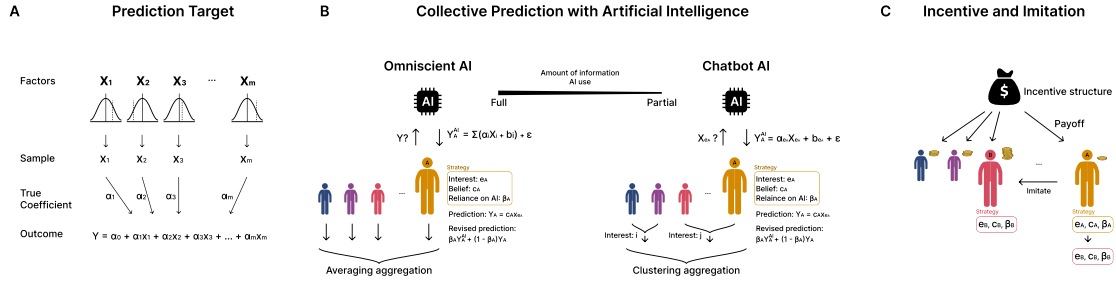


Figure 1: **A schematic diagram of the model for individual and collective prediction in the presence of personal AI assistants.** (A) The prediction target is modeled as a linear combination of causal factors. (B) Each player is characterized by three strategies: interest, belief, and reliance on AI. Players make predictions based on their interests and beliefs and subsequently revise them with AI assistance. We consider two types of AI, Chatbot AI and Omniscient AI, which differ in the amount of information available for generating predictions. (C) Players receive payoffs according to the incentive structure, and the population evolves through imitation of peers with higher payoffs.

We begin by extending previously established models of CI, in which a group of players participates in a prediction task (Figure 1): each player makes an individual prediction based on partial observations of the underlying causal factors, and the group forms a collective prediction by aggregating these individual predictions (Hong et al., 2012; Mann and Helbing, 2017; Wang et al., 2025). Specifically, the prediction target, Y , is modeled as a linear combination of M causal factors x_i , with fixed coefficients α_i : $Y = \sum_{i=1}^M \alpha_i x_i + \alpha_0$, where each factor is drawn from independent normal distribution (Figure 1A). Then, each individual A observes a single factor of their interest x_{e_A} and makes a prediction based on their own belief, c_A : $Y_A = c_A x_{e_A}$ (Figure 1B). We assume each individual can then independently consult their AI assistants and revise their prediction based on their reliance on AI, β_A :

92 $Y_A^{\text{re}} = \beta_A Y_A^{\text{AI}} + (1 - \beta_A) Y_A$ (Figure 1B). The revised predictions are then aggregated
 93 using two alternative aggregation rules, averaging and clustering, to form collective
 94 prediction.

95 To model AI, we assume that AI has already been trained on similar tasks and is
 96 able to make predictions based on partial or full observations of causal factors (Figure
 97 1B). Every time a player consults AI, it will make a new prediction Y_A^{AI} based on
 98 its internal parameters, which can include factor-specific bias (b_i) and stochastic
 99 error ϵ . Here, we consider two extreme types of AI response, which differ in the
 100 amount of information AI uses when making predictions. In one extreme, the AI
 101 can observe all causal factors and use all information to make predictions, regardless
 102 of the user’s interest: $Y_A^{\text{AI}} = \sum_{i=0}^M (\alpha_i x_i + b_i) + \epsilon$. In the other extreme, AI works
 103 like a chatbot, where the AI predicts the outcome based only on the user’s interest:
 104 $Y_A^{\text{AI}} = \alpha_{e_A} x_{e_A} + b_{e_A} + \epsilon$. Hereafter, we refer to these two types as ‘Omniscient AI’
 105 and ‘Chatbot AI’, respectively.

106 We assume that appropriate aggregation rule for collective prediction depends
 107 on the type of AI. When Chatbot AI is used, each revised prediction depends on a
 108 single factor. Thus, we use clustering aggregation, under which revised predictions
 109 are first averaged among individuals who are interested in the same factor, and
 110 then summed across factors. When Omniscient AI is used, however, each revised
 111 prediction contains information about all causal factors, in which case clustering
 112 these predictions by interest would be inappropriate. Instead, we use averaging
 113 aggregation, under which revised individual predictions are averaged directly to form
 114 the collective prediction.

115 Each generation consists of multiple round of predictions, and each player is re-
 116 warded a payoff according to the incentive structure (Figure 1C). For simplicity,
 117 we assume that a sufficiently large number of prediction rounds occurs within each
 118 generation, allowing us to use the expected payoff value rather than computing the
 119 payoffs for every round. We consider two incentive structures from previous work,
 120 “feedback” and “niche-expert”, which have been shown to promote near-perfect col-
 121 lective accuracy in the absence of AI (Wang et al., 2025). The feedback structure
 122 gives higher payoff to individuals who can push the collective prediction toward the
 123 true target, regardless of their individual predictive accuracy: $\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}})$. In
 124 contrast, the niche-expert structure gives higher payoff to individuals who have both
 125 better predictive accuracy and a minority interest: $\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2$. Here, ρ_{e_A}
 126 denotes the proportion of players interested in the same factor, thereby allowing the
 127 niche-expert structure to preserve interest diversity. We note collective intelligence
 128 fails to emerge under the combination of averaging aggregation with niche-expert
 129 structure in the absence of AI (Wang et al., 2025).

130 Through time, the group evolves by allowing each individual to imitate peers
 131 who received higher payoffs (Figure 1C). Specifically, after each generation, two
 132 individuals are randomly sampled, and one imitates all three strategies (the factor of
 133 interest, factor belief, and reliance on AI) of the other with a probability determined
 134 by the difference in their expected payoffs (Materials and Methods). Individual

135 predictions are aggregated into a collective prediction $\hat{Y}$ based on the aggregation
 136 rules. Collective accuracy is defined as the reduction in prediction error achieved
 137 by the collective prediction relative to a baseline prediction α_0 (Wang et al., 2025):
 138 $1 - \mathbb{E}[(Y - \hat{Y}^{\text{re}})^2] / \mathbb{E}[(Y - \alpha_0)^2]$. Hereafter, we say that CI emerges when the collective
 139 accuracy reaches a near perfect level.

140 This model now allows us to explore the impact of AI on CI depending on the
 141 type of AI and the incentive structure. Here, we begin by comparing the emergence
 142 of CI with and without AI and subsequently investigate a how different incentive
 143 structures affect the emergence of CI in the presence of AI.

144 AI can either promote or prevent the emergence of 145 CI

146 To understand the impact of AI, we first compare collective dynamics with and
 147 without AI. First, consistent with Wang et al. (2025), simple imitation rules and
 148 incentive structures allow collective intelligence to emerge in the absence of AI (Fig.
 149 2A,C,D). One exception is the use of simple averaging with niche-expert incentive
 150 structures, under which the collective accuracy of the population remains near zero.
 151 This is because enhancing individual accuracy causes individuals to predict close to
 152 the true coefficient of their interest, but simply averaging these predictions does not
 153 necessarily produce an accurate estimate of the full prediction target (Fig. S1).

154 We find that AI usage can have heterogeneous outcomes depending on its type.
 155 For example, the use of Omniscient AI can either preserve CI under the feedback
 156 incentive structure (Fig. 2A) or promote higher collective accuracy under the niche-
 157 expert incentive structure (Fig. 2B). These results are robust across different levels
 158 of AI accuracy and can be shown analytically by assuming a sufficiently large group
 159 of players (Fig. S2; see SI Evolutionary dynamics section). **[SWP: need to get**
 160 **back.]**

161 In contrast, the use of Chatbot AI prevents the emergence of CI, thereby resulting
 162 in lower levels of collective accuracy (Fig. 2C, D). Under the feedback incentive
 163 structure, the use of Chatbot AI leads to a loss of diversity in the population's
 164 interest, making it impossible to account for effect of some causal factors (Fig. S2,
 165 S3). This causes an increase in collective variance and stochastic decrease in collective
 166 accuracy (Fig. 2C; Fig. S3). We note that the loss of interest diversity may arise
 167 even in the absence of AI in finite populations (Fig. S3), but the use of Chatbot
 168 AI makes the group more vulnerable to such loss (Fig. S4). Under the niche-
 169 expert incentive structure, the population becomes overly reliant on AI, causing the
 170 collective prediction to inherit the AI's bias (Fig. S3). This leads to an increase in
 171 collective bias and a decrease in collective accuracy (Fig. 2; Fig. S3). Interestingly,
 172 we find that the niche-expert structure can prevent the use of AI under specific
 173 assumptions about the relationship between the true coefficient and AI bias (Fig.
 174 S2). In this case, the population reduces **[JS: I think it is difficult to say that it**

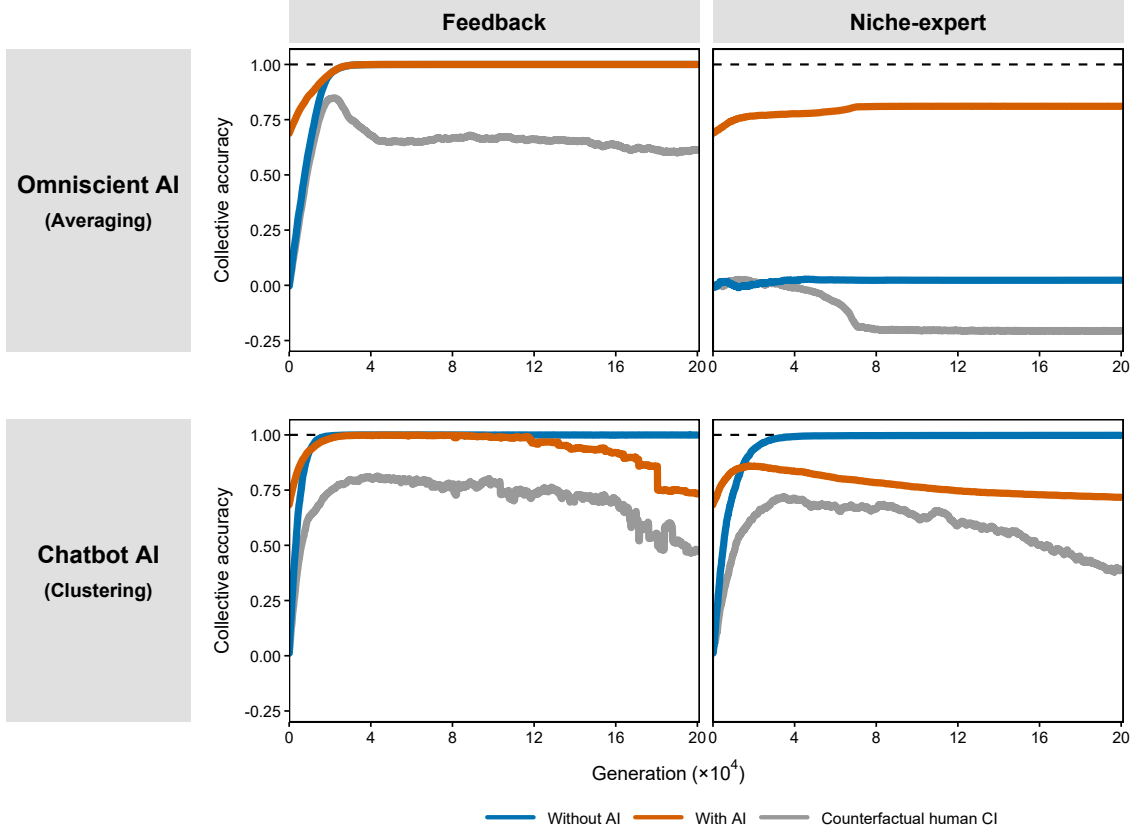


Figure 2: **Comparison of collective accuracy with and without AI.** Trajectories of collective accuracy when the group is assisted by the Omniscient AI (A,B) and Chatbot AI (C,D). Here, feedback (A,C) and niche-expert (B,D) incentive structures are considered. Blue lines represent trajectories of collective accuracy without AI assistance. Orange lines represent trajectories of collective accuracy with AI assistance. Gray lines represent trajectories of counterfactual collective accuracy, computed from players’ unrevised predictions, indicating the group’s predictive accuracy if AI assistance were unavailable at the given generation. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A, B), $c_A \sim \mathcal{N}(0, 5^2)$ for (C, D), and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.36$ and $\tau = 0.3$.

completely excludes AI, because reliance on AI does not approach zero. This may be due to the limited population size.] the reliance on AI assistance, allowing the CI to emerge eventually. However, this phenomenon can substantially delay the emergence of CI (Fig. S5).

To investigate whether the group can retain its predictive ability when AI assistance suddenly becomes unavailable at any given time, we quantify counterfactual collective accuracy based on unrevised predictions. Across all scenarios, we find that

the counterfactual collective accuracy is lower than the collective accuracy of the corresponding population (Fig. 2).

In summary, an omniscient AI can promote CI, but AI operating with limited information may prevent the emergence of CI. Moreover, reliance on AI can limit inherent ability of the population to make accurate predictions on their own, raising concerns about future AI usage. These findings further underscore the need for incentive structures that can mitigate the potential risks associated with AI assistance.

Directly incentivizing or penalizing AI use can additionally harm CI

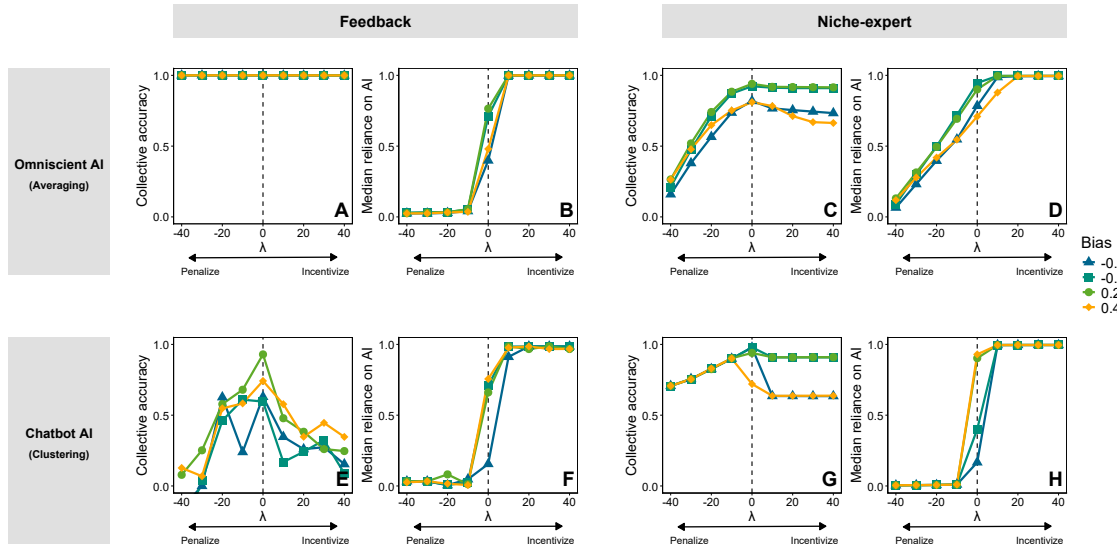


Figure 3: Collective accuracy under the influence of direct intervention on AI use. Collective accuracy and median reliance on AI under varying strengths of intervention (λ) across different assumptions about AI types (omniscient AI, A–D and chatbot AI, E–H) and incentive structures (Feedback, A,B,E,F, and Niche expert, C,D,G,H). Positive λ represents an incentive for AI use, whereas negative λ represents a penalty for AI use. In each panel, the vertical dashed line indicates the original incentive structure ($\lambda = 0$). Four AI bias levels are shown: $b_i = \pm 0.2$, representing higher-accuracy AI, and $b_i = \pm 0.4$, representing lower-accuracy AI. Collective accuracy (A,C,E,G) and median reliance on AI (B,D,F,H) are averaged over generations 190,000–200,000. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A–D), $c_A \sim \mathcal{N}(0, 5^2)$ for (E–H), and $\beta_A \sim U(0, 1)$. AI parameter: $\tau = 0.3$.

Given the strong influence of AI on collective outcomes, a straightforward inter-

vention is to directly incentivize or penalize the use of AI. To investigate whether such direct incentives or penalties can help the emergence of CI, we modify the two incentive structures by adding a payoff term proportional to individual reliance on AI:

$$\text{Feedback/Niche-expert} + \lambda\beta_A \quad (1)$$

where λ represent the strength of intervention ($\lambda > 0$) or penalty ($\lambda < 0$), and β_A represent individual reliance on AI. [SWP: maybe we need to call incentive pi earlier on...] [SWP: also good to define beta again here as a reminder.]

Directly incentivizing or penalizing AI use can further reduce the collective accuracy (Fig. 3C, E, G), except when Omniscient AI is used under the feedback structure; in the latter case, the population retains collective intelligence regardless of the intervention strength (Fig. 3A). The reduction in collective accuracy arises because incentivizing or penalizing AI usage causes individuals to imitate others based on their AI usage rather than their predictive accuracy. Thus, stronger interventions can push the group to always [JS: wholly?] ($\beta = 1$) or never ($\beta = 0$) rely on AI (Fig. 3B, D, F, H) but these scenarios do not necessarily lead to better outcomes (Fig. 3D, F, H). When AI bias is relatively large, a weak penalty can lead to higher collective accuracy, but this represents a narrow parameter regime (Figure 3G).

Overall, these results indicate that simply incentivizing or penalizing the AI, independent of individuals' performance, use may exacerbate rather than resolve the risk of AI on CI. Thus, we also considered an incentive structure where interventions for AI usage affects the payoffs multiplicatively, thus giving higher payoffs to individuals who rely more (or less) AI and makes better prediction. [SWP: Supp.] [JS: Should I add the multiplicative intervention to the SI? I remember that we didn't reach a clear conclusion during our last meeting about whether to include it.]

Balanced incentives mitigate AI-induced failures of CI

Neither the original incentive structures nor simple interventions targeting AI use are sufficient to promote CI. What incentive structure, then, would work in the presence of AI? Here, we propose a "balanced" incentive structure, modeled as the average of the feedback and niche expert incentives.

The core idea behind the balanced structure is to maintain interest diversity and prevent collective bias. On one hand, the feedback incentive structure is prone to the loss of interest diversity but can prevent collective bias by incentivizing players whose predictions lie opposite to the collective prediction. On the other hand, the niche-expert structure can maintain interest diversity because it directly incentivizes minority interests. However the niche-expert structure cannot prevent collective bias because, within the same factor of interest, individuals are rewarded only according to their individual predictive performance, which in turn homogenize the predictions.

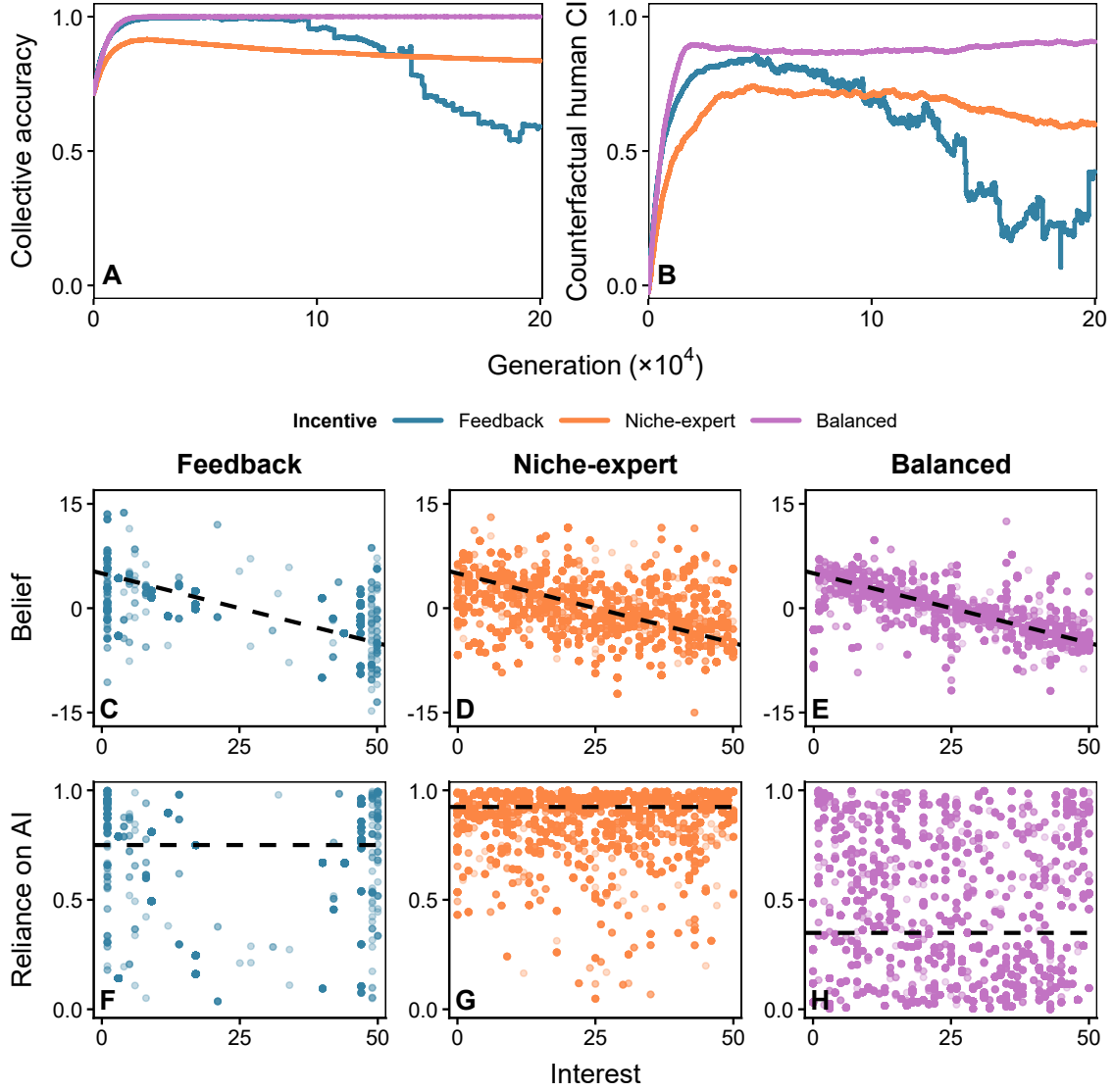


Figure 4: **Balanced structure can promote collective intelligence** Trajectories of collective accuracy (A) and counterfactual human CI (B) under three incentive structures (feedback, niche-expert, and balanced) with Chatbot AI. Distributions of individual interest and belief (C-E) and of interest and reliance on AI (F-H) after 200,000 generations. Each point represents an individual. Dashed lines in panels C-E represent the true coefficients (α_i) of the prediction target. Dashed horizontal lines in panels F-H indicate the median reliance on AI. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i = 5 - 0.2i$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.4$ and $\tau = 0.3$.

230 Therefore, when the two incentive structures are combined, the population achieves
 231 CI by maintaining interest diversity while limiting collective bias (Fig. 4A, S6). Un-

der the balanced structure, individuals remain more evenly distributed across causal factors, and their beliefs lie closer to the true coefficients than under either incentive structure alone (Fig. 4C–E). The balanced structure also produces higher counterfactual collective accuracy than the feedback or niche-expert structure (Fig. 4B, S6). These patterns are robust to how the two incentive structures are weighted when combined (Fig. S7), and are driven by the emergent property of the balanced structure, under which the population increases its reliance on AI as AI becomes more accurate (Fig S8).

[SWP: *may need some robustness check rather than a case study...*]

Discussion

The ongoing race towards the development of “better” AIs continues to pose questions about the societal benefits and risks of AI uses (Bender et al., 2021; Burton et al., 2024; Cui and Yasseri, 2024). Here, we used mathematical models of evolutionary dynamics to investigate how AI uses and incentive structures jointly affect the emergence of CI, which is defined as the population’s ability to make accurate predictions. We find that an “omniscient” AI model, which incorporates information about all relevant causal factors, can improve collective accuracy, whereas a “chatbot” AI model, whose outcome depends on the user’s interest, can impair it. These contrasting outcomes arise because AI assistance affects not only individual predictions, but also the population’s interest diversity, beliefs, and reliance on AI. We also show that directly incentivizing or penalizing AI uses, independent of individual’s performances, can further limit the collective accuracy of the population. Instead, a balanced incentive structure can maintain CI by promoting diversity and limiting bias.

Our analysis reveals that AI can prevent the emergence of CI through two different mechanisms. First, AI use can cause individuals to concentrate on a subset of the factors that determine the prediction target, thereby reducing diversity in their interests. This result is broadly consistent with recent empirical studies showing that AI assistance can improve individual performance but limit the diversity of outputs, in domains ranging from scientific research (Hao et al., 2026) to creative writing (Doshi and Hauser, 2024). Second, over-reliance on AI can cause the population to inherit inherent biases in AI’s predictions (Fügener et al., 2021; Vicente and Matute, 2023), causing the prediction accuracy to decrease eventually. This mechanism can prevent the emergence of CI even when the interest diversity is maintained because biases in AI predictions can be amplified when individual predictions are aggregated. Thus, preserving the diversity of ideas will not be sufficient to promote CI. Instead, AI uses need to be calibrated to prevent systematic AI biases from becoming collective biases (Bućinca et al., 2021).

Our analyses of counterfactual collective accuracy also raises an important concern for future AI uses: while AI uses may improve the apparent performance of

272 individuals and, in some cases, the population as a whole, it can simultaneously re-
 273 duce the population’s ability to make accurate predictions without AI. These results
 274 echo findings from a recent field experiment, which demonstrated that students,
 275 who seemingly performed better in solving mathematics problems under AI assis-
 276 tance, performed worse on their own than those who did not have access (Bastani
 277 et al., 2025). These findings highlight the need to evaluate AI tools not only by
 278 the performance gains they provide during use, but also by whether individuals and
 279 populations retain the ability to perform without them (Melumad and Yun, 2025;
 280 Ke et al., 2026). Making this distinction clear will be particularly important for
 281 education, where AI may help students complete a task without helping them learn
 282 how to do it on their own (Bastani et al., 2025; Melumad and Yun, 2025).

283 As generative AIs and large language models are already generating substantial
 284 benefits, many organizations, both industrial and academic, are increasingly seek-
 285 ing to improve their collective performance by regulating AI uses (Thorp, 2023;
 286 Bousquette, 2026; Chawla, 2026; Keary, 2026). Such regulations may take direct
 287 forms, including AI-use policies or token-based leaderboards, or may operate indi-
 288 rectly through other incentives, such as productivity rewards or the social stigma
 289 associated with AI use (Reif et al., 2025). Our model suggests that directly incen-
 290 tivizing or penalizing AI use is unlikely to induce appropriately calibrated reliance on
 291 AI or to promote CI. Incentivizing AI use could lead to overreliance, causing groups
 292 to propagate AI biases, whereas penalizing AI use could lead to underreliance, pre-
 293 venting groups from using AI even when it performs well in the relevant domain.
 294 Our result instead shows that appropriate AI use and accurate CI can be promoted
 295 by focusing on the traditional properties of groups that have been emphasized for
 296 decades: maintaining diversity and preventing collective bias, rather than on AI use
 297 itself (Hong and Page, 2004; Lorenz et al., 2011; Mann and Helbing, 2017).

298 There are several limitations to our analyses. First, our model allows for social
 299 learning among humans, but AI’s inherent ability remains fixed throughout the evo-
 300 lutionary process. In reality, AI systems can also learn from human-generated data
 301 (Ouyang et al., 2022). Such bidirectional learning increase the similarity between
 302 human and AI errors and therefore further reduce the group performance (?). Sec-
 303 ond, we model AI errors using fixed, factor-specific additive biases and independent
 304 random errors. This assumption allows random errors to be reduced through ag-
 305 gregation, while systematic biases can still persist in the collective prediction. In
 306 reality, AI errors may have more complex structures, including biased estimates of
 307 factor effects and correlated errors across individuals, factors, or repeated interac-
 308 tions. Further empirical and theoretical work is therefore needed to understand how
 309 these error structures affect the emergence of CI. Finally, following established mod-
 310 els of CI, we assume that the prediction target is a linear combination of independent
 311 causal factors. This assumption does not hold for many real-world tasks, where causal
 312 factors may be correlated or interact nonlinearly. Future studies should therefore in-
 313 vestigate how nonlinear relationships between causal factors and prediction targets
 314 affect the interaction between AI assistance and the emergence of CI.

315 The rapid adoption of personalized AI assistants is changing how individuals
 316 learn, work, and make decisions (Tessler et al., 2024; Glickman and Sharot, 2025;
 317 Sajadieh et al., 2026; Bick et al., 2026; Ke et al., 2026). While the immediate ben-
 318 efits and risks of these systems have been evaluated at the individual level Noy and
 319 Zhang (2023); Bastani et al. (2025), our study raises a broader concern that their
 320 widespread AI use may reshape collective intelligence of human society. Our results
 321 further underline the importance of designing reward structures that preserve minor-
 322 ity expertise and prevent AI biases from persisting across the population. A detailed
 323 empirical understanding of how AI assistance reshapes diversity, expertise, and re-
 324 liance at the population level will therefore be critical for predicting the dynamics of
 325 future of human collective intelligence.

326 Methods

327 Prediction target

328 Following previous models of CI, we model the prediction target as a linear combi-
 329 nation of M independent causal factors (Hong et al., 2012; Mann and Helbing, 2017;
 330 Wang et al., 2025):

$$Y = \alpha_0 + \alpha_1 X_1 + \cdots + \alpha_M X_M, \quad (2)$$

331 where the coefficient α_i determines the magnitude and direction of the effect of
 332 factor i , and α_0 represents a global intercept. Each causal factor X_i is modeled as an
 333 independent Gaussian random variable with mean zero and standard deviation σ_i .

334 Players

335 We consider a well-mixed population $\mathcal{N}$ consisting of N players engaged in a novel
 336 prediction task. In the absence of AI assistance, the strategy of player A is char-
 337 acterized by two variables: interest e_A and belief c_A . Player A observes the causal
 338 factor X_{e_A} and makes a prediction by multiplying the observed factor value by their
 339 belief:

$$Y_A = c_A X_{e_A}, \quad (3)$$

340 where e_A can take an integer value from 0 to M . When $e_A = 0$, the player does
 341 not observe any causal factor and instead makes a constant prediction based solely
 342 on their belief. We refer to this as a stubborn strategy because the player’s predic-
 343 tion does not respond to the realized values of the causal factors. For notational
 344 convenience, we assume that $X_0 = 1$.

345 When the group is assisted by AI, each player’s strategy additionally includes
 346 reliance on AI, denoted by β_A . We assume that each player consults the AI inde-
 347 pendently and revises their prediction by taking a weighted average of their own
 348 prediction and the AI-generated prediction:

$$Y_A^{\text{re}} = \beta_A Y_A^{\text{AI}} + (1 - \beta_A) Y_A, \quad (4)$$

where β_A ranges from 0 to 1. A value of $\beta_A = 1$ indicates complete reliance on AI, whereas $\beta_A = 0$ indicates that the player does not use the AI prediction.

AI models

We assume that the AI model has been trained on similar tasks and is therefore capable of making predictions based on causal factors. However, because the model is imperfect, its outputs are subject to systematic additive bias and stochastic error. The additive bias associated with factor i is denoted by b_i , and the stochastic error is denoted by ϵ . The AI is neither trained nor updated during the simulation, and its biases therefore remain fixed throughout the simulation. For simplicity, we assume $b_i = b$ for the main analysis for all i . The model formulation nevertheless allows biases to differ among factors.

We assume that the stochastic error follows a Gaussian distribution with mean zero and standard deviation τ . Each time a player consults the AI, an independent error is generated. Errors are therefore independent across players and across repeated consultations by the same player. Consequently, stochastic AI errors can be attenuated when independently generated predictions are aggregated across players. We therefore hold τ constant and vary the magnitude of the additive bias to manipulate the level of AI accuracy during our analyses.

We consider two extreme types of AI that differ in the amount of information available for prediction. The first is termed the Omniscient AI, which can observe all M causal factors and incorporate them into its prediction regardless of the user's interest. The prediction generated by the Omniscient AI for player A is expressed as follows:

$$Y_A^{\text{AI}} = \alpha_0 + b_0 + \alpha_1 X_1 + b_1 + \cdots + \alpha_M X_M + b_M + \epsilon. \quad (5)$$

The second is termed the Chatbot AI, which can use only the single causal factor X_{e_A} associated with the player's interest. The prediction generated by the Chatbot AI for player A is expressed as follows:

$$Y_A^{\text{AI}} = \alpha_{e_A} X_{e_A} + b_{e_A} + \epsilon. \quad (6)$$

Aggregation rules

The revised individual predictions are aggregated to form a collective prediction. We use two aggregation rules proposed in previous research (Wang et al., 2025), depending on the type of AI used by the group. First, when the group is assisted by the Omniscient AI, the AI component of each revised prediction contains information about all M factors, even though the individual's own prediction is based on only one factor. Each revised prediction can therefore be treated as an estimate of the entire target. Accordingly, we calculate the collective prediction by taking the simple average of the revised individual predictions:

$$\hat{Y}^{\text{re}} = \frac{Y_1^{\text{re}} + Y_2^{\text{re}} + \cdots + Y_n^{\text{re}}}{N}. \quad (7)$$

384 Second, when the group is assisted by the Chatbot AI, both the player’s initial
 385 prediction and the AI’s prediction are based on the same causal factor. Each revised
 386 prediction therefore contains information about only one factor and can be treated
 387 as an estimate of that factor’s contribution to the target. We consequently use
 388 clustering aggregation.

389 Clustering aggregation is performed by grouping individuals according to their
 390 interests and calculating the mean prediction within each group. The collective
 391 prediction is then obtained by summing the mean predictions of all nonempty groups:

$$\hat{Y}^{\text{re}} = \sum_{i:n_i>0} \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} Y_A^{\text{re}}, \quad (8)$$

392 where $\mathcal{A}_i$ represents the set of players interested in factor i , and $n_i = |\mathcal{A}_i|$ represents
 393 the number of players in that set.

394 When the group is not assisted by AI, collective predictions are computed from the
 395 unrevised individual predictions throughout the simulation. Both simple averaging
 396 and clustering can be used in the absence of AI. For fair comparison, we use the
 397 same aggregation rule as in the corresponding AI-assisted condition.

398 Incentive structures

399 Players receive payoffs based on the underlying incentive structure. We consider five
 400 incentive structures: feedback structure, niche-expert structure, direct intervention
 401 on AI use of these two structures, and a balanced structure.

402 The feedback structure does not directly reward individuals according to their
 403 own prediction accuracy. Instead, it rewards predictions that are aligned with the
 404 residual error of the collective prediction and therefore have the potential to move
 405 the collective prediction toward the true value. Given player A ’s revised prediction
 406 Y_A^{re} and the collective prediction $\hat{Y}^{\text{re}}$, the payoff is expressed as

$$\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}). \quad (9)$$

407 The niche-expert structure assigns higher payoffs to individuals who make accu-
 408 rate predictions while having an underrepresented interest. Specifically, the squared
 409 prediction error of player A is weighted by $\rho_{e_A} = n_{e_A}/n$, the proportion of players
 410 interested in the same factor:

$$\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2. \quad (10)$$

411 Previous research has shown that these two incentive structures can produce near
 412 perfect collective accuracy through social learning, except when the niche-expert
 413 incentive is combined with simple averaging (Wang et al., 2025). They are therefore
 414 used as baseline structures for examining how AI assistance affects the emergence of
 415 CI.

416 To examine the effects of directly encouraging or discouraging AI use, we mod-
 417 ify the feedback and niche-expert structures by adding a term proportional to the
 418 player’s reliance on AI:

$$\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) + \lambda\beta_A \quad (11)$$

$$\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2 + \lambda\beta_A. \quad (12)$$

419 Here, λ represents the strength of the direct incentive or penalty applied to AI use. A
 420 positive value of λ rewards AI reliance, whereas a negative value penalizes it. When
 421 $\lambda = 0$, no direct incentive or penalty is applied, and the payoff functions reduce to
 422 the original feedback and niche-expert structures.

423 The balanced structure is defined as the arithmetic mean of the feedback and
 424 niche-expert payoff components:

$$\pi_A = \frac{Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) - \rho_{e_A}(Y_A^{\text{re}} - Y)^2}{2}. \quad (13)$$

425 Simulations

426 All simulations are conducted using an agent-based model with 50 causal factors
 427 ($M = 50$) and 10,000 players ($n = 10,000$). The parameters defining the prediction
 428 target are sampled at the beginning of the simulations and subsequently held fixed.

429 Coefficients α_i are sampled from a uniform distribution on $[-5, 5]$, except in the
 430 simulation presented in Figure 4, in which α_i is set to $5 - 0.2i$ for visualization.
 431 Standard deviations σ_i are sampled from a uniform distribution on $[0, 3]$.

432 Players’ initial strategies are also sampled from predefined probability distribu-
 433 tions. The distribution of individual belief c_A depends on the aggregation rule. Under
 434 simple averaging, each factor’s contribution is scaled by the proportion of players in-
 435 terested in that factor. We therefore sample beliefs from a Gaussian distribution
 436 with mean zero and standard deviation 100, following the scaling used in previous
 437 work (Wang et al., 2025). Under clustering aggregation, beliefs are sampled from a
 438 Gaussian distribution with mean zero and standard deviation 5. Because predictions
 439 are averaged separately within each interest group and then summed, no additional
 440 scaling of c_A is required. The initial interest of each player is sampled uniformly
 441 from the integers between 0 and 50, and reliance on AI was sampled from a uniform
 442 distribution on $[0, 1]$.

443 We assume that prediction events occur on a substantially faster timescale than
 444 social learning. In each generation, we therefore calculate the expected payoff $\mathbb{E}[\pi_A]$
 445 of each player analytically rather than explicitly sampling prediction events. The
 446 expectation is taken over the distributions of the causal factors $X_1, \dots, X_M$ and the
 447 stochastic AI error ϵ , conditional on the current population strategies and the fixed
 448 target parameters. Closed form expression of expected payoffs are derived for all
 449 incentive structures and are presented in the Supplementary Methods.

Each generation consists of an expected payoff calculation followed by one opportunity for pairwise social learning. Two distinct players, A and B , are randomly sampled, and player A imitated all three strategies of player B with probability

$$\frac{1}{1 + \exp(s(\mathbb{E}[\pi_A] - \mathbb{E}[\pi_B]))}, \quad (14)$$

where s represents the strength of social learning and was set to 50 in all simulations. We introduce a mutation rate μ , such that with probability μ , player A does not imitate player B but instead redrew all strategy components from the same distributions used to initialize the population. The mutation rate is set to zero in all simulations except those presented in Supplementary Fig. S4 [SWP: *need to check*].

Unless otherwise specified, each simulation is run for 200,000 generations. When a result is represented by a single summary value rather than as a temporal trajectory, the metric was averaged over the final 10,000 generations.

To evaluate the dynamics of collective prediction, we define six summary metrics: collective accuracy, collective bias, collective variance, counterfactual collective accuracy, interest diversity, and median reliance on AI. At each population state, expectations and variances are evaluated analytically over the distributions of the causal factors and AI errors rather than estimated from sampled prediction events. Collective bias and collective variance constitute the bias–variance decomposition of the group level collective prediction error.

Collective accuracy is normalized relative to the baseline prediction α_0 . A negative value indicates that the collective prediction performs worse than the baseline. Counterfactual collective accuracy is evaluated by setting $\beta_A = 0$ while retaining the evolved interests and beliefs of the AI-assisted population. In calculating interest diversity, we use the convention that $\rho_i \ln \rho_i = 0$ when $\rho_i = 0$ (Table 1).

Data and code availability

All data are stored in a publicly available GitHub repository (<https://github.com/parklab-snu/collective-intelligence-with-AI>).

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Contributions

Jaeseung Kim: Conceptualization, Formal Analysis, Investigation, Methodology, Software, Visualization, Writing - Original Draft Preparation, Writing - Review &

Table 1: Evaluation metrics used to analyze collective prediction dynamics.

Metric	Definition	Interpretation
Collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y}^{\text{re}})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy of the revised collective prediction relative to the baseline prediction α_0 .
Collective bias	$\left(\mathbb{E}[Y - \hat{Y}^{\text{re}}]\right)^2$	Squared expected deviation of the revised collective prediction from the true target.
Collective variance	$\text{Var}(Y - \hat{Y}^{\text{re}})$	Variability of the revised collective prediction error across prediction events.
Counterfactual collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy obtained after removing AI use from the current population while retaining its evolved interests and beliefs.
Interest diversity	$\exp\left(-\sum_{i=0}^M \rho_i \ln \rho_i\right)$	Effective number of interests represented in the population.
Median reliance on AI	$\text{median}_{A \in N}(\beta_A)$	Median reliance on AI in the population.

483 Editing. Sang Woo Park: Conceptualization, Investigation, Methodology, Supervi-
484 sion, Writing - Original Draft Preparation, Writing - Review & Editing.

485 **Competing Interests**

486 The authors declare no competing interests.

Supplementary Text

Here, we derive the expected payoffs and collective accuracy used in the simulations. We consider Omniscient AI and Chatbot AI separately because they contain different amounts of information and therefore require different aggregation rules. We then analyze the corresponding evolutionary dynamics under a deterministic large-population approximation.

1 Omniscient AI

We first consider Omniscient AI, which incorporates all causal factors regardless of the player's interest. We derive individual and collective predictions before calculating the expected payoffs under each incentive structure.

1.1 Model

Because Omniscient AI observes all M causal factors, its prediction for player A is given by

$$Y_A^{AI} = \alpha_0 + b_0 + \alpha_1 X_1 + b_1 + \cdots + \alpha_M X_M + b_M + \epsilon, \quad (S1)$$

where b_i represent factor-specific biases and ϵ represents stochastic error, following a Gaussian distribution with mean zero and standard deviation τ . A new stochastic error is generated independently for each player and prediction event.

Let $B = \sum_{i=0}^M b_i$ denote the total additive bias. Because the true prediction target is $Y = \alpha_0 + \sum_{i=1}^M \alpha_i X_i$, the AI prediction can be written as

$$Y^{AI} = Y + B + \epsilon. \quad (S2)$$

In contrast to the AI prediction, player A 's initial prediction depends only on the causal factor associated with their interest. Their initial and revised predictions are therefore given by:

$$Y_A = c_A X_{e_A} \quad (S3)$$

$$Y_A^{re} = \beta_A Y^{AI} + (1 - \beta_A) Y_A \quad (S4)$$

represents player A 's interest, c_A represents their belief about the effect of the corresponding factor, and β_A represents their reliance on AI. As in the main text, we set $X_0 = 1$, such that a player with $e_A = 0$ makes a constant initial prediction.

511 1.2 Aggregation

512 Because the AI component of each revised prediction contains information about all
513 M causal factors, we use simple averaging to obtain the collective prediction:

$$\hat{Y}^{re} = \frac{1}{N} \sum_{j=1}^N Y_j^{re} \quad (S5)$$

$$= \frac{1}{N} \sum_{j=1}^N \{\beta_j Y^{AI} + (1 - \beta_j) Y_j\} \quad (S6)$$

$$= \frac{1}{N} \sum_{j=1}^N \{\beta_j (Y + B + \epsilon) + (1 - \beta_j) c_j X_{e_j}\} \quad (S7)$$

514 To separate the contributions of AI and the players' initial predictions, we define the
515 population's mean reliance on AI, the stochastic AI error remaining after aggregation,
516 and the aggregate contribution of initial predictions associated with each factor:

$$\bar{\beta} = \frac{1}{N} \sum_{j=1}^N \beta_j \quad (S8)$$

$$\bar{\eta} = \frac{1}{N} \sum_{j=1}^N \beta_j \epsilon \quad (S9)$$

$$\gamma_i = \frac{1}{N} \sum_{A \in \mathcal{A}_i} (1 - \beta_A) c_A \quad (S10)$$

517 Here, $\bar{\beta}$ represents the mean reliance on AI, $\bar{\eta}$ represents the stochastic AI error
518 remaining after averaging, and γ_i represents the contribution of players' initial pre-
519 dictions to factor i . Using these definitions, the revised collective prediction can be
520 written as follows:

$$\hat{Y}^{re} = \bar{\beta}(Y + B) + \sum_{i=0}^M \gamma_i X_i + \bar{\eta} \quad (S11)$$

521 Thus, the collective prediction consists of an AI-derived component, the contributions
522 of the players' initial predictions, and the stochastic AI error that remains after
523 aggregation.

524 1.3 Expected payoffs

525 We next derive the expected payoff under each incentive structure. Expectations are
526 taken over the distributions of the causal factors and stochastic AI errors, conditional
527 on the current population strategies and the fixed parameters of the prediction target.

528 Feedback

529 Under the feedback structure, a player's payoff depends on the alignment between
 530 their revised prediction and the residual error of the collective prediction. To express
 531 this residual, we first define the remaining error in the intercept and in the coefficient
 532 of each causal factor:

$$\delta_0 = \alpha_0 - (\bar{\beta}(\alpha_0 + B) + \gamma_0), \quad (\text{S12})$$

$$\delta_i = \alpha_i - (\bar{\beta}\alpha_i + \gamma_i), \quad (i \neq 0), \quad (\text{S13})$$

$$Y - \hat{Y}^{re} = \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta}, \quad (\text{S14})$$

533 where δ_0 represents the remaining error in the intercept, whereas δ_i represents the
 534 remaining error associated with factor i . Because the causal factors and stochastic AI
 535 errors are independent and have zero mean, cross terms vanish when the expectation
 536 is taken. Substituting the revised individual prediction and collective residual into
 537 the feedback payoff gives

$$\mathbb{E}(\pi_A) = \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] \quad (\text{S15})$$

$$= \mathbb{E} \left[\left\{ \beta_A \alpha_0 + \beta_A \sum_{i=1}^M \alpha_i X_i + \beta_A B + \beta_A \epsilon + (1 - \beta_A) c_A X_{e_A} \right\} \right] \quad (\text{S16})$$

$$\times \left[\left\{ \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \right\} \right] \quad (\text{S17})$$

$$= \beta_A \alpha_0 \delta_0 + \beta_A \sum_{i=1}^M \delta_i \alpha_i \sigma_i^2 + \beta_A B \delta_0 - \frac{1}{N} \beta_A^2 \tau^2 + (1 - \beta_A) c_A \delta_{e_A} \sigma_{e_A}^2 \quad (\text{S18})$$

$$= \beta_A \left\{ (\alpha_0 + B) \delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2 \right\} + (1 - \beta_A) c_A \delta_{e_A} \sigma_{e_A}^2 - \frac{\beta_A^2 \tau^2}{N} \quad (\text{S19})$$

538 The first term represents the alignment between the AI-derived prediction and the
 539 collective residual, whereas the second term represents the contribution of the player's
 540 initial prediction. The final term arises because the stochastic error in player A 's AI
 541 prediction also contributes to the collective prediction. This finite-population effect
 542 decreases as population size increases.

543 Niche-expert

544 Under the niche-expert structure, the payoff is the negative squared error of the
 545 revised individual prediction, weighted by the frequency of the player's interest.
 546 Because $X_0 = 1$, a player with $e_A = 0$ makes a constant initial prediction. In this

case, the expected payoff is:

$$\mathbb{E}(\pi_A) = -\mathbb{E}[\rho_0(Y_A^{re} - Y)^2] \quad (\text{S20})$$

$$= -\mathbb{E}[\rho_0\{(\beta_A - 1)Y + \beta_A B + \beta_A \epsilon + (1 - \beta_A)c_A\}^2] \quad (\text{S21})$$

$$= -\mathbb{E}\left[\rho_0\left\{(\beta_A - 1)(\alpha_0 - c_A) + \beta_A B + (\beta_A - 1)\sum_{i=1}^M \alpha_i X_i + \beta_A \epsilon\right\}^2\right] \quad (\text{S22})$$

$$= -\rho_0\left\{(1 - \beta_A)^2(\alpha_0 - c_A)^2 - 2\beta_A(1 - \beta_A)(\alpha_0 - c_A)B + \beta_A^2 B^2\right. \quad (\text{S23})$$

$$\left. + (1 - \beta_A)^2 \sum_{i \neq 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2\right\} \quad (\text{S24})$$

For a player with $e_A \neq 0$, the initial prediction instead depends on the observed value of factor X_{e_A} . The expected payoff is therefore:

$$\mathbb{E}(\pi_A) = -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2] \quad (\text{S25})$$

$$= -\mathbb{E}[\rho_{e_A}\{(\beta_A - 1)Y + \beta_A B + \beta_A \epsilon + (1 - \beta_A)c_A X_{e_A}\}^2] \quad (\text{S26})$$

$$= -\mathbb{E}[\rho_{e_A}\{(\beta_A - 1)\alpha_0 + \beta_A B$$

$$+ (\beta_A - 1)(\alpha_{e_A} - c_A)X_{e_A} + (\beta_A - 1)\sum_{i \neq e_A, 0} \alpha_i X_i + \beta_A \epsilon\}^2] \quad (\text{S28})$$

$$= -\rho_{e_A}\left\{(1 - \beta_A)^2 \alpha_0^2 - 2\beta_A(1 - \beta_A)\alpha_0 B\right. \quad (\text{S29})$$

$$+ \beta_A^2 B^2 + (1 - \beta_A)^2(\alpha_{e_A} - c_A)^2 \sigma_{e_A}^2$$

$$\left. + (1 - \beta_A)^2 \sum_{i \neq e_A, 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2\right\} \quad (\text{S31})$$

These expressions separate the contributions of the player's belief, the unobserved causal factors, systematic AI bias, and stochastic AI error. Increasing reliance on AI reduces the contribution of errors in the player's initial prediction but increases their exposure to AI bias and stochastic error.

Incentivizing or penalizing AI use

Direct incentives add the term $\lambda\beta_A$ to either the feedback or niche-expert payoff. Because this term is fixed conditional on the player's strategy, its expectation is simply $\lambda\beta_A$. The expected payoffs therefore become

$$\mathbb{E}(\pi_{A, \text{Feedback}}) = \mathbb{E}\left[Y_A^{re}(Y - \hat{Y}^{re}) + \lambda\beta_A\right] = \mathbb{E}\left[Y_A^{re}(Y - \hat{Y}^{re})\right] + \lambda\beta_A, \quad (\text{S32})$$

$$\mathbb{E}(\pi_{A, \text{Niche-expert}}) = -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2 + \lambda\beta_A] = -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2] + \lambda\beta_A. \quad (\text{S33})$$

Thus, a positive value of λ increases the payoff of strategies with greater reliance on AI, whereas a negative value favors strategies with lower reliance.

560 1.4 Expected accuracy

561 Using the residual defined above, the mean squared error of the revised collective
562 prediction is:

$$\mathbb{E}[(Y - \alpha_0)^2] = \mathbb{E} \left[\left(\alpha_0 + \sum_{i=1}^M \alpha_i X_i - \alpha_0 \right)^2 \right] = \sum_{i=1}^M \alpha_i^2 \sigma_i^2 \quad (\text{S34})$$

$$\mathbb{E}[(Y - \hat{Y}^{re})^2] = \mathbb{E} \left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i - \hat{\eta} \right)^2 \right] = \delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \frac{\tau^2}{N^2} \sum_{j=1}^N \beta_j^2 \quad (\text{S35})$$

563 The expected collective accuracy is therefore

$$R = 1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]} \quad (\text{S36})$$

$$= 1 - \frac{\delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \frac{\tau^2}{N^2} \sum_{j=1}^N \beta_j^2}{\sum_{i=1}^M \alpha_i^2 \sigma_i^2} \quad (\text{S37})$$

564 The collective prediction error consists of the squared error in the intercept, the re-
565 maining error associated with each causal factor, and the stochastic AI error remain-
566 ing after averaging across players. Counterfactual collective accuracy is obtained
567 from the same expression by setting $\beta_A = 0$ for all players while retaining their
568 evolved interests and beliefs.

569 2 Chatbot AI

570 Unlike Omniscient AI, Chatbot AI uses only the causal factor associated with each
571 player's interest. This causes the information contained in AI predictions to depend
572 directly on the distribution of interests in the population. Here, we derive the collec-
573 tive prediction, expected payoffs, and expected collective accuracy under this form
574 of AI assistance.

575 2.1 Model

576 The prediction generated by Chatbot AI for player A depends on the causal factor
577 associated with the player's interest:

$$Y_A^{AI} = \alpha_{e_A} X_{e_A} + b_{e_A} + \epsilon, \quad (\text{S38})$$

578 where b_{e_A} represents factor-specific additive bias, and ϵ represents stochastic error.
579 The stochastic error follows a Gaussian distribution with mean zero and standard
580 deviation τ and is generated independently across players and prediction events. The

581 player's initial and revised predictions are given by Player A's initial and revised
 582 predictions can be then written as follows:

$$Y_A = c_A X_{e_A} \quad (\text{S39})$$

$$Y_A^{re} = \beta_A Y_A^{AI} + (1 - \beta_A) Y_A \quad (\text{S40})$$

583 Here, c_A represents player A's belief about the effect of their focal factor, e_A repre-
 584 sents their interest, and β_A represents their reliance on AI. Thus, the revised predic-
 585 tion remains specific to the single factor associated with the player's interest.

586 2.2 Aggregation

587 Because both the player's initial prediction and the Chatbot AI prediction contain
 588 information about the same causal factor, we use clustering aggregation. Revised
 589 predictions are first averaged among players interested in the same factor and then
 590 summed across represented interests:

$$\hat{Y}^{re} = \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} Y_A^{re} \quad (\text{S41})$$

$$= \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \{ \beta_A Y_A^{AI} + (1 - \beta_A) Y_A \} \quad (\text{S42})$$

$$= \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} [\{ \beta_A \alpha_{e_A} + (1 - \beta_A) c_A \} X_{e_A} + \beta_A b_{e_A} + \beta_A \epsilon], \quad (\text{S43})$$

591 where $\mathcal{A}_i$ is the set of players interested in factor i , and $n_i = |\mathcal{A}_i|$ is the number of
 592 players in that set.

593 To simplify the collective prediction, we define the effective coefficient, weighted
 594 AI bias, and stochastic error associated with each interest group:

$$\tilde{c}_i = \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \{ \beta_A \alpha_{e_A} + (1 - \beta_A) c_A \}, \quad (\text{S44})$$

$$\tilde{B}_i = \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \beta_A b_{e_A}, \quad (\text{S45})$$

$$\bar{B} = \sum_{i=0}^M \tilde{B}_i, \quad (\text{S46})$$

$$\tilde{\eta}_i = \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \beta_A \epsilon, \quad (\text{S47})$$

$$\bar{\eta} = \sum_{i=0}^M \tilde{\eta}_i, \quad (\text{S48})$$

where $\tilde{c}_i$ represents the mean effective coefficient for factor i after AI assistance, whereas $\tilde{B}_i$ and $\tilde{\eta}_i$ represent its systematic and stochastic error components. Thus, the collective prediction can be written as:

$$\hat{Y}^{re} = \sum_{i \in \mathcal{I}} \tilde{c}_i X_i + \bar{B} + \bar{\eta}, \quad (\text{S49})$$

where $\mathcal{I} = i \in 0, \dots, M > 0$ denotes the set of represented interests.

2.3 Expected payoffs

We next derive the expected payoff under each incentive structure. Expectations are taken over the causal factors and stochastic AI errors, conditional on the current population strategies and the fixed parameters of the prediction target.

Feedback

Under the feedback structure, players are rewarded according to whether their revised predictions reduce the error in the collective prediction. We therefore define the remaining errors in the intercept and factor-specific coefficients as

$$\delta_0 = \alpha_0 - \tilde{c}_0 - \bar{B}, \quad (\text{S50})$$

$$\delta_i = \alpha_i - \tilde{c}_i, \quad (i \neq 0), \quad (\text{S51})$$

$$Y - \hat{Y}^{re} = \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta}, \quad (\text{S52})$$

where δ_0 includes the remaining error in the intercept and the total additive AI bias, whereas δ_i measures the remaining error in the coefficient of factor i . Because the causal factors and stochastic errors are independent and have zero mean, substituting the revised individual prediction into the feedback payoff gives

$$\mathbb{E}(\pi_A) = \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] \quad (\text{S53})$$

$$= \mathbb{E}[\{(\beta_A \alpha_{e_A} + (1 - \beta_A) c_A) X_{e_A} + \beta_A b_{e_A} + \beta_A \epsilon\} \quad (\text{S54})$$

$$\times \left\{ \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \right\}] \quad (\text{S55})$$

$$= \{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} \delta_{e_A} \sigma_{e_A}^2 + \beta_A b_{e_A} \delta_0 - \frac{1}{n_{e_A}} \beta_A^2 \tau^2 \quad (\text{S56})$$

Thus, the expected feedback payoff depends on the remaining collective error associated with the player's interest, the contribution of factor-specific AI bias to the intercept error, and the player's contribution to the stochastic error of their interest group.

615 Niche-expert

616 Under the niche-expert structure, the negative squared error of each revised indi-
 617 vidual prediction is weighted by the frequency of the player's interest. We consider
 618 $e_A = 0$ separately because these players make constant predictions rather than factor-
 619 dependent predictions. When $e_A = 0$, the expected payoff is

$$\mathbb{E}(\pi_A) = -\mathbb{E}[\rho_0(Y_A^{re} - Y)^2] \quad (\text{S57})$$

$$= -\mathbb{E} \left[\rho_0 \left[\left\{ \beta_A \alpha_0 + (1 - \beta_A) c_A \right\} + \beta_A b_0 + \beta_A \epsilon - \alpha_0 - \sum_{i=1}^M \alpha_i X_i \right]^2 \right] \quad (\text{S58})$$

$$= -\rho_0 \left\{ (1 - \beta_A)^2 (c_A - \alpha_0)^2 + 2(1 - \beta_A)(c_A - \alpha_0)\beta_A b_0 + \beta_A^2 b_0^2 \right. \quad (\text{S59})$$

$$\left. + \sum_{i \neq 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2 \right\} \quad (\text{S60})$$

620 When $e_A \neq 0$, the player's prediction varies with their focal causal factor. The
 621 expected payoff is then

$$\mathbb{E}(\pi_A) = -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2] \quad (\text{S61})$$

$$= -\mathbb{E} [\rho_{e_A} [\{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} X_{e_A} \quad (\text{S62})$$

$$+ \beta_A b_{e_A} + \beta_A \epsilon - \alpha_0 - \sum_{i=1}^M \alpha_i X_i]^2] \quad (\text{S63})$$

$$= -\rho_{e_A} \left\{ (1 - \beta_A)^2 (c_A - \alpha_{e_A})^2 \sigma_{e_A}^2 + \sum_{i \neq e_A, 0} \alpha_i^2 \sigma_i^2 \right. \quad (\text{S64})$$

$$\left. + \alpha_0^2 + \beta_A^2 b_{e_A}^2 - 2\alpha_0 \beta_A b_{e_A} + \beta_A^2 \tau^2 \right\} \quad (\text{S65})$$

622 In both cases, the expected squared error consists of the mismatch between the
 623 player's belief and the true coefficient, contributions from causal factors omitted
 624 from the individual prediction, factor-specific AI bias, and stochastic AI error. The
 625 niche-expert weight ρ_{e_A} scales these errors according to the frequency of the player's
 626 interest.

627 Incentivizing or penalizing AI use

628 Directly incentivizing or penalizing AI use adds the term $\lambda \beta_A$ to the player's payoff.
 629 Because this term is fixed conditional on the player's strategy, it can be taken outside

the expectation. The expected payoffs are therefore

$$\mathbb{E}(\pi_{A,Feedback}) = \mathbb{E} \left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) + \lambda\beta_A \right] \quad (\text{S66})$$

$$= \mathbb{E} \left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) \right] + \lambda\beta_A, \quad (\text{S67})$$

$$\mathbb{E}(\pi_{A,Niche-expert}) = -\mathbb{E} \left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2 + \lambda\beta_A \right] \quad (\text{S68})$$

$$= -\mathbb{E} \left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2 \right] + \lambda\beta_A. \quad (\text{S69})$$

Balanced

Under the balanced structure, the payoff corresponds to the arithmetic mean of the feedback and niche-expert payoff components:

$$\mathbb{E}(\pi_A) = \mathbb{E} \left[\frac{Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) - \rho_{e_A}(Y_A^{\text{re}} - Y)^2}{2} \right] \quad (\text{S70})$$

$$= \frac{\mathbb{E} \left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) \right] - \mathbb{E} \left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2 \right]}{2} \quad (\text{S71})$$

The expected payoff under the balanced structure can therefore be calculated directly from the feedback and niche-expert payoffs derived above.

2.4 Expected accuracy

Finally, we derive the expected collective accuracy under clustering aggregation. The intercept α_0 serves as the baseline prediction. Because the causal factors are independent and have mean zero, its expected squared error is

$$\mathbb{E}[(Y - \alpha_0)^2] = \mathbb{E} \left[\left(\alpha_0 + \sum_{i=1}^M \alpha_i X_i - \alpha_0 \right)^2 \right] = \sum_{i=1}^M \alpha_i^2 \sigma_i^2 \quad (\text{S72})$$

Using the expression for the collective residual above, the expected squared error of the revised collective prediction is

$$\mathbb{E}[(Y - \hat{Y})^2] = \mathbb{E} \left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i - \hat{\eta} \right)^2 \right] = \delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \sum_{i=0}^M \frac{\tau^2}{n_i^2} \sum_{A \in \mathcal{A}_i} \beta_A^2 \quad (\text{S73})$$

The first two terms represent squared errors in the intercept and factor-specific coefficients. The final term represents the stochastic AI error remaining after predictions are averaged within each interest group. Thus, stochastic error has a larger effect when an interest is represented by fewer individuals. Normalizing this error relative

646 to the baseline gives:

$$R = 1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]} \quad (\text{S74})$$

$$= 1 - \frac{\delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \sum_{i=0}^M \frac{\tau^2}{n_i^2} \sum_{A \in \mathcal{A}_i} \beta_A^2}{\sum_{i=1}^M \alpha_i^2 \sigma_i^2} \quad (\text{S75})$$

647 Here, $R = 1$ represents perfect collective prediction, $R = 0$ represents performance
648 equal to the baseline, and $R < 0$ represents performance worse than the baseline.

649 3 Evolutionary dynamics

650 For sufficiently large population ($N \rightarrow \infty$), we can approximate the deterministic
651 evolutionary dynamics using the replicator equation under both weak and strong
652 selections. We focus on the feedback and niche-expert structures, for which the long-
653 term dynamics can be characterized analytically. The direct interventions on AI use
654 and the balanced structure are examined through simulations.

655 Let λ denote a strategy, p_λ its frequency, and $u_\lambda = \mathbb{E}(\pi_\lambda)$ its expected payoff.
656 Under replicator dynamics, the frequency of strategy λ changes according to

$$\frac{dp_\lambda}{dt} = p_\lambda(u_\lambda - \bar{u}), \quad (\text{S76})$$

657 where $\bar{u} = \sum_\lambda p_\lambda u_\lambda$. Therefore, at an internal stationary state, all strategies
658 with positive frequency have the same expected payoff. A boundary state can also
659 be stationary because a strategy with exactly zero frequency cannot increase under
660 the replicator equation. Thus, stability of a boundary state needs to be considered
661 separately by examining invasion by a rare strategy.

662 3.1 Omniscient AI

663 3.1.1 Feedback

664 When the population size approaches infinity, the effect of stochastic error on the
665 collective prediction vanishes ($\bar{\eta} \rightarrow 0$) and the effect of stochastic error in the ex-
666 pected payoff, $-\beta^2 \tau^2 / N$, also vanishes. Following prior research (Wang et al., 2025),
667 we define the Lyapunov function as the expected error of collective prediction

$$V = \mathbb{E} \left[(Y - \hat{Y}^{re})^2 \right]. \quad (\text{S77})$$

668 The function V is non-negative and equals zero when the collective prediction is
669 identical to the target. To examine how V changes under the evolutionary dynamics,
670 we take its time derivative:

$$\begin{aligned}
\frac{dV}{dt} &= -2\mathbb{E} \left[(Y - \hat{Y}^{re}) \frac{d\hat{Y}^{re}}{dt} \right] \\
&= -2\mathbb{E} \left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i \right) \left\{ \frac{d\bar{\beta}}{dt} (Y + B) + \sum_{i=0}^M \frac{d\gamma_i}{dt} X_i \right\} \right] \\
&= -2 \left[\left\{ (\alpha_0 + B) \delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2 \right\} \frac{d\bar{\beta}}{dt} + \delta_0 \frac{d\gamma_0}{dt} + \sum_{i=1}^M \delta_i \sigma_i^2 \frac{d\gamma_i}{dt} \right].
\end{aligned}$$

671 For notational convenience, we define

$$K = (\alpha_0 + B) \delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2. \quad (\text{S78})$$

672 Although K changes with the population state, it is common to all strategies at a
673 given state and can therefore be treated as a constant with respect to the strategy
674 index λ . We also define

$$G_e = \begin{cases} \delta_0, & e = 0, \\ \delta_e \sigma_e^2, & e \neq 0. \end{cases} \quad (\text{S79})$$

675 By the definitions of $\bar{\beta}$ and γ_i and the replicator equation,

$$\frac{d\bar{\beta}}{dt} = \sum_{\lambda} p_{\lambda} (u_{\lambda} - \bar{u}) \beta_{\lambda} = \text{Cov}(\beta, u). \quad (\text{S80})$$

676 Similarly, the terms involving γ_i can be combined as

$$\delta_0 \frac{d\gamma_0}{dt} + \sum_{i=1}^M \delta_i \sigma_i^2 \frac{d\gamma_i}{dt} = \text{Cov}((1 - \beta) c G_e, u). \quad (\text{S81})$$

677 Therefore,

$$\begin{aligned}
\frac{dV}{dt} &= -2 \{ K \text{Cov}(\beta, u) + \text{Cov}((1 - \beta) c G_e, u) \} \\
&= -2 \text{Cov}(K\beta + (1 - \beta) c G_e, u).
\end{aligned}$$

678 For the feedback structure in the limit $N \rightarrow \infty$, the expected payoff of strategy
679 λ is

$$u_{\lambda} = K \beta_{\lambda} + (1 - \beta_{\lambda}) c_{\lambda} G_{e_{\lambda}}. \quad (\text{S82})$$

680 Which is same with Thus,

$$\frac{dV}{dt} = -2 \text{Cov}(u, u) = -2 \text{Var}(u) \leq 0. \quad (\text{S83})$$

Therefore, the expected error of collective prediction cannot increase during the evolutionary process and decreases whenever strategies differ in their expected payoffs. The derivative becomes zero when the variance of expected payoffs is zero, $\text{Var}(u) = 0$. This condition corresponds to equal expected payoffs among the strategies present in the population. A boundary state in which the entire population consists of a single strategy can also satisfy $\text{Var}(u) = 0$ even when $V > 0$. However, such a state is not evolutionarily stable under the feedback structure. If the collective prediction differs from the target Y , a nonzero residual remains. By the definition of the feedback payoff, a novel strategy whose prediction aligns with the direction of this residual obtains a higher expected payoff than the resident strategy and can therefore invade the population. Thus, an imperfect boundary equilibrium with $V > 0$ is unstable to the introduction of novel predictions.

Consequently, the evolutionary dynamics continue to reduce the collective prediction error until the population reaches the stable state with zero residual, $V = 0$. At this state, the collective prediction equals the target, $\hat{Y}^{re} = Y$, and the collective accuracy reaches $R = 1$.

3.1.2 Niche-expert

Unlike the stochastic error in the collective prediction, stochastic error in the niche-expert payoff is evaluated at the individual level and does not vanish as population size increases. Therefore, we retain $\tau > 0$ in the following derivation.

We first consider strategies having the same interest by optimizing the expected payoff with respect to c and β . For a nonzero interest e , the expected payoff is strictly concave with respect to c for $\beta < 1$ and $\sigma_e > 0$, and is maximized at

$$c_e^* = \alpha_e. \quad (\text{S84})$$

After substituting $c_e^* = \alpha_e$, the payoff is strictly concave with respect to β . Its unconstrained stationary point is

$$\tilde{\beta}_e = \frac{\alpha_0^2 + \alpha_0 B + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2}{(\alpha_0 + B)^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S85})$$

Because β is constrained between 0 and 1, the optimal β_e^* is $\tilde{\beta}_e$ when $0 < \tilde{\beta}_e < 1$, and otherwise lies at the corresponding boundary, $\beta_e^* = 0$ or $\beta_e^* = 1$. Thus, the boundary cases do not require a separate evolutionary argument; they only change the optimal prediction and its minimum expected squared error. When $\beta_e^* = 1$, the revised prediction is independent of c_e , so different values of c_e are selectively neutral and can be treated as a single payoff-equivalent strategy class.

Let L_e denote the minimum expected individual error, $\mathbb{E}[(Y_A^{re} - Y)^2]$, within interest e . For an internal optimum,

$$L_e = \frac{B^2 \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2 \left(\alpha_0^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 \right)}{(\alpha_0 + B)^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S86})$$

714 If the optimum is located at $\beta_e^* = 0$ or $\beta_e^* = 1$, L_e is instead obtained by evaluating
 715 the expected squared prediction error at the corresponding boundary.

716 For interest 0, optimization with respect to c gives

$$c_0^*(\beta) = \alpha_0 - \frac{\beta}{1 - \beta} B. \quad (\text{S87})$$

717 Substituting this into the expected payoff gives a strictly concave function of β ,
 718 with the unique optimum

$$\beta_0^* = \frac{\sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S88})$$

719 Therefore,

$$c_0^* = \alpha_0 - \frac{B}{\tau^2} \sum_{j=1}^M \alpha_j^2 \sigma_j^2, \quad (\text{S89})$$

720 and

$$L_0 = \frac{\tau^2 \sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S90})$$

721 We denote the optimal strategy, or the payoff-equivalent strategy class when
 722 $\beta_e^* = 1$, by S_e . A state containing a suboptimal strategy within an interest can
 723 be stationary if the optimal strategy is absent, but it is unstable against invasion
 724 because the optimal strategy has a strictly higher payoff at the same ρ_e . Therefore,
 725 a stable state can contain only S_e within each represented interest.

726 We next consider the frequencies of the $M + 1$ optimal strategies. Let ρ_e denote
 727 the frequency of S_e . Its expected payoff is

$$u_e = -\rho_e L_e. \quad (\text{S91})$$

728 Assuming $L_e > 0$ for all interests, the equal-payoff condition of the replicator
 729 equation at an internal stationary state gives

$$\rho_0^* L_0 = \rho_1^* L_1 = \cdots = \rho_M^* L_M. \quad (\text{S92})$$

730 Together with $\sum_{e=0}^M \rho_e^* = 1$, this uniquely determines

$$\rho_e^* = \frac{D}{L_e}, \quad D = \left(\sum_{j=0}^M \frac{1}{L_j} \right)^{-1}. \quad (\text{S93})$$

731 This internal stationary state is locally stable. At equilibrium, all represented
 732 interests have the same expected payoff, with $\rho_e^* L_e = D$. If the frequency of one
 733 interest increases slightly above its equilibrium value, its payoff decreases because
 734 the niche-expert payoff is proportional to $-\rho_e L_e$. Conversely, if the frequency of
 735 an interest decreases below its equilibrium value, its payoff increases. The replicator
 736 dynamics therefore generate a restoring force that reduces small deviations in interest
 737 frequencies and drives the population back toward the internal stationary state.

738 If one or more interests are absent, the resulting boundary state can be stationary
 739 under the replicator equation, but it is unstable against invasion. The payoff of a rare
 740 absent interest approaches zero as $\rho_e \rightarrow 0$, whereas the mean payoff of the resident
 741 population is negative when $L_e > 0$, so the rare interest increases when introduced
 742 at sufficiently small frequency.

743 Thus, there exists a unique stable full-support equilibrium in terms of the optimal
 744 prediction strategies and their frequencies. Whether the constrained optimum has
 745 $\beta_e^* = 0$, $0 < \beta_e^* < 1$, or $\beta_e^* = 1$ does not change the frequency argument, because
 746 these cases differ only through the resulting value of L_e . When $\beta_e^* = 1$, the strategy
 747 is selectively neutral with respect to c_e , but its prediction, payoff, and equilibrium
 748 interest frequency remain uniquely determined.

749 This equilibrium also allows a direct comparison with the expected accuracy of
 750 the AI. Using Y^{AI} itself without aggregation as the prediction gives

$$\mathbb{E} [(Y - Y^{\text{AI}})^2] = B^2 + \tau^2. \quad (\text{S94})$$

751 For every interest, complete reliance on AI, $\beta = 1$, is a feasible strategy. There-
 752 fore, because L_e is the minimum individual expected squared prediction error within
 753 interest e ,

$$L_e \leq B^2 + \tau^2. \quad (\text{S95})$$

754 Moreover, for interest 0,

$$L_0 = \frac{\tau^2 \sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2} < \tau^2 \leq B^2 + \tau^2. \quad (\text{S96})$$

755 Because the stable equilibrium has $\rho_0^* > 0$,

$$\sum_{e=0}^M \rho_e^* L_e < B^2 + \tau^2. \quad (\text{S97})$$

756 The collective prediction is the average of individual revised predictions. By
 757 convexity of the squared error,

$$\mathbb{E} [(Y - \hat{Y}^{re})^2] \leq \sum_{e=0}^M \rho_e^* L_e. \quad (\text{S98})$$

758 Therefore,

$$\mathbb{E} \left[(Y - \hat{Y}^{re})^2 \right] < \mathbb{E} \left[(Y - Y^{AI})^2 \right]. \quad (\text{S99})$$

759 Because the collective accuracy and AI accuracy use the same normalization by
 760 $\mathbb{E}[(Y - \alpha_0)^2]$, the stationary collective accuracy under the niche-expert structure is
 761 strictly greater than the expected accuracy of the AI prediction.

762 Although the niche-expert structure improves collective prediction relative to the
 763 AI, the stationary collective accuracy does not reach one when $\tau > 0$. For a nonzero
 764 factor i , the residual coefficient of the stationary collective prediction can be written
 765 as

$$\delta_i^* = \alpha_i \left\{ 1 - \bar{\beta}^* - \rho_i^* (1 - \beta_i^*) \right\} = \alpha_i \sum_{e \neq i} \rho_e^* (1 - \beta_e^*). \quad (\text{S100})$$

766 At the stable equilibrium, $\rho_0^* > 0$. In addition, $\tau > 0$ implies $\beta_0^* < 1$. Therefore,
 767 for every factor with $\alpha_i \neq 0$,

$$\delta_i^* \neq 0. \quad (\text{S101})$$

768 Because $\sum_{i=1}^M \alpha_i^2 \sigma_i^2 > 0$, at least one factor contributes positive prediction error.
 769 Hence,

$$\mathbb{E} \left[(Y - \hat{Y}^{re})^2 \right] > 0, \quad (\text{S102})$$

770 and consequently the stationary collective accuracy satisfies $R^* < 1$.

Supplementary Figure

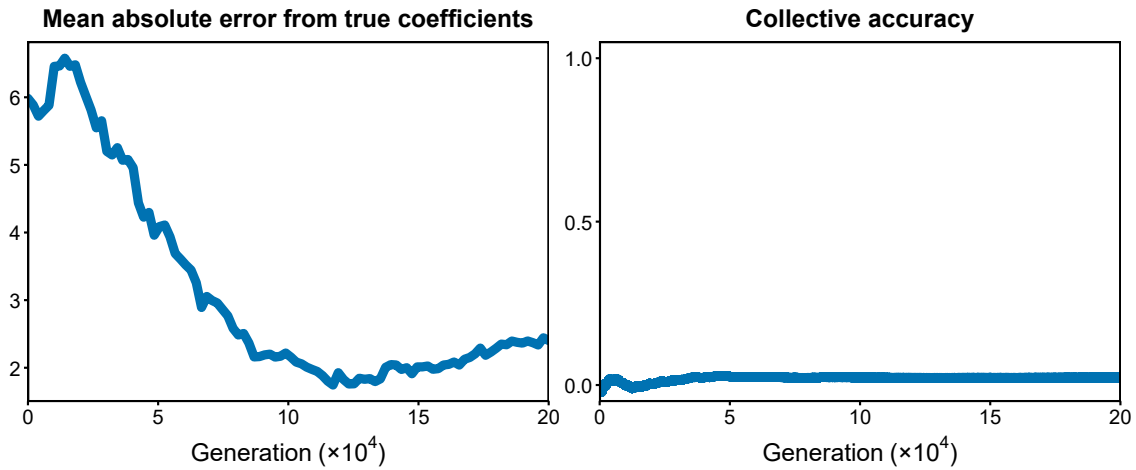


Figure S1: **Niche-expert structure with averaging aggregation cannot promote CI without AI.** Trajectories of the mean absolute error from true coefficients and collective accuracy under the niche-expert structure with averaging aggregation in the absence of AI. The mean absolute error from true coefficients is calculated by first taking, for each interest, the absolute difference between the true coefficient and the mean belief of individuals sharing that interest, and then averaging these differences across all $m + 1$ interests. The mean absolute error decreases across generations, indicating that the mean beliefs converge toward their corresponding true coefficients. However, collective accuracy remains near zero, indicating that simply averaging individual predictions does not yield accurate collective predictions. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$.

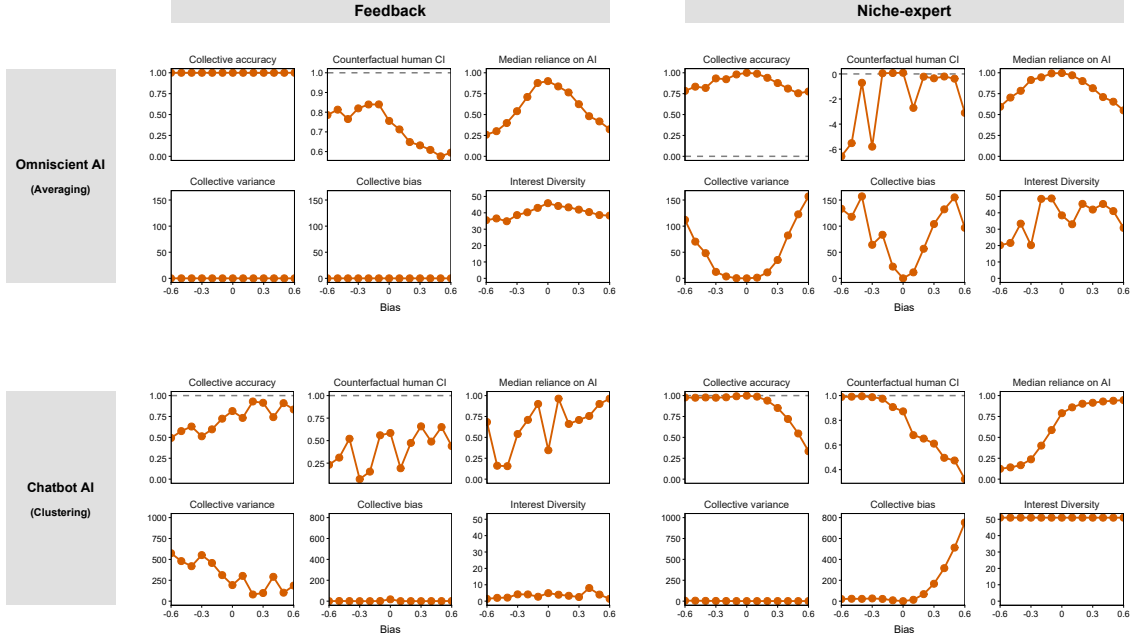


Figure S2: **Evaluation metrics across various levels of AI bias.** Evaluation metrics across varying levels of AI bias under different assumptions about AI types (omniscient AI, and chatbot AI) and incentive structures (feedback, niche-expert). Larger absolute AI bias indicates a less accurate AI model. Across the range of AI bias considered, AI accuracy varies from approximately 0.24 to nearly 1 when the AI predicts the target independently. Dashed horizontal lines represent the values of the evaluation metrics when the group is not assisted by AI. Collective accuracy represents the predictive performance of the group, and counterfactual human CI represents the group’s predictive accuracy if AI assistance were unavailable at the given generation. Interest diversity is defined as Hill-Shannon diversity of interest within the population, median reliance on AI represents the median value of reliance on AI (β). Collective bias represents systematic deviation of the collective prediction from the true target, collective variance represents variability of the collective prediction error across prediction events. All the evaluation metrics are averaged over generations 190,000–200,000. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A, B), $c_A \sim \mathcal{N}(0, 5^2)$ for (C, D), and $\beta_A \sim U(0, 1)$. AI parameters: $\tau = 0.3$.

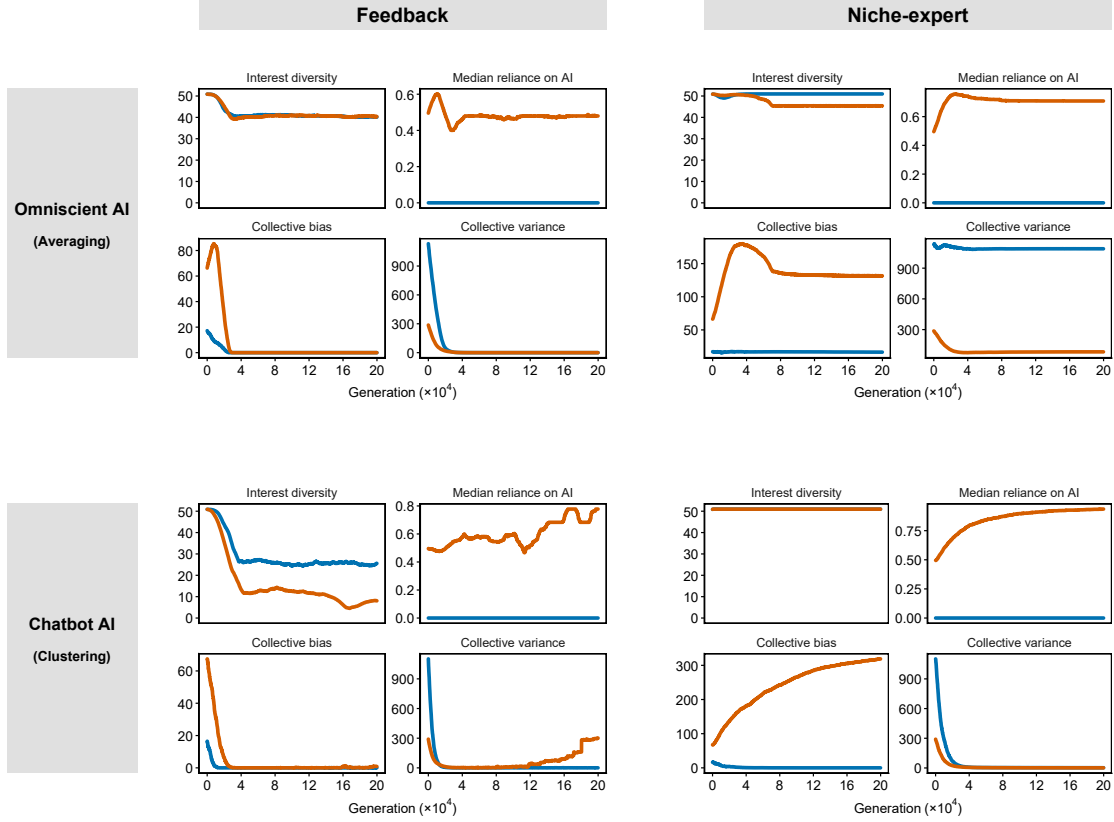


Figure S3: **Comparison of evaluation metrics with and without AI.** Trajectories of evaluation metrics when the group is assisted by the Omniscient AI and Chatbot AI. Here, feedback and niche-expert incentive structures are considered. Blue lines represent trajectories of evaluation metrics without AI assistance. Orange lines represent trajectories of evaluation metrics with AI assistance. Interest diversity is define as hill-shannon diversity of interest within population, median reliance on AI represent population median of reliance on AI (β), collective bias represent systematic deviation of the collective prediction from the true target, collective variance repressent variability of the collective prediction error across prediction events. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A, B), $c_A \sim \mathcal{N}(0, 5^2)$ for (C, D), and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.36$ and $\tau = 0.3$.

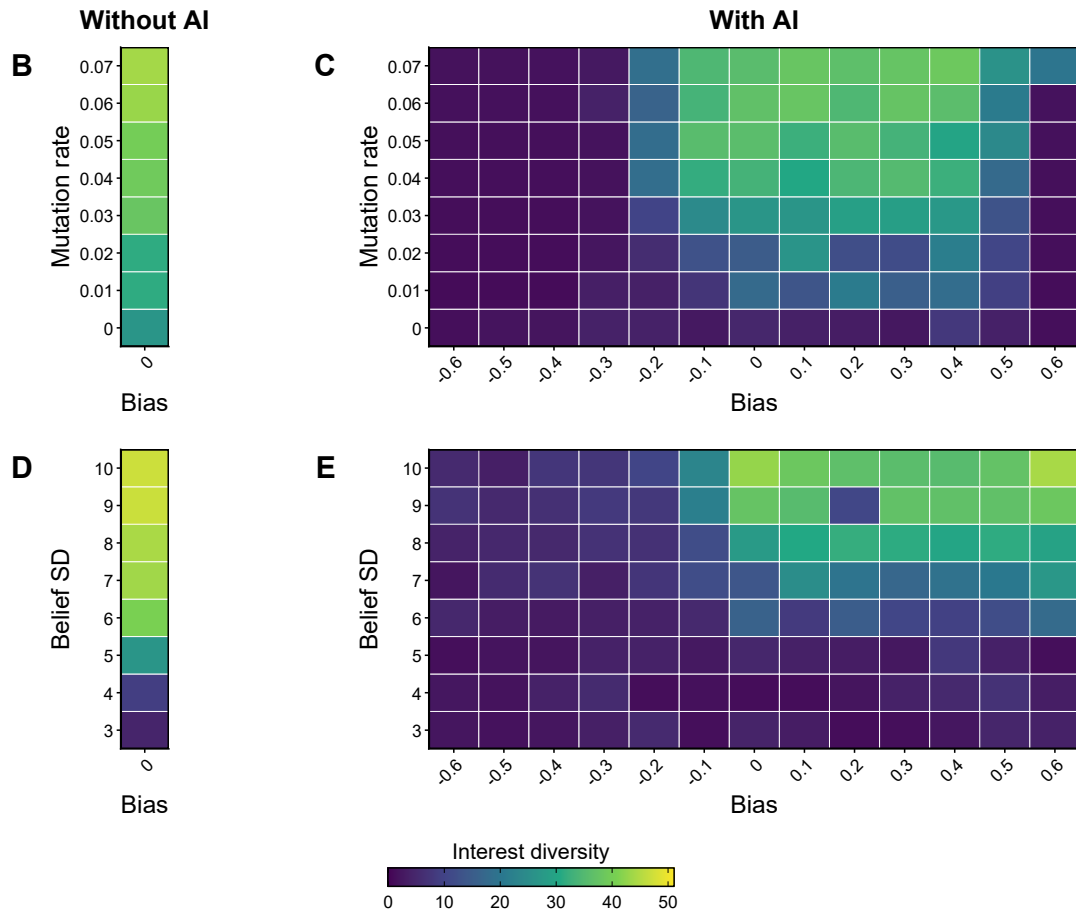
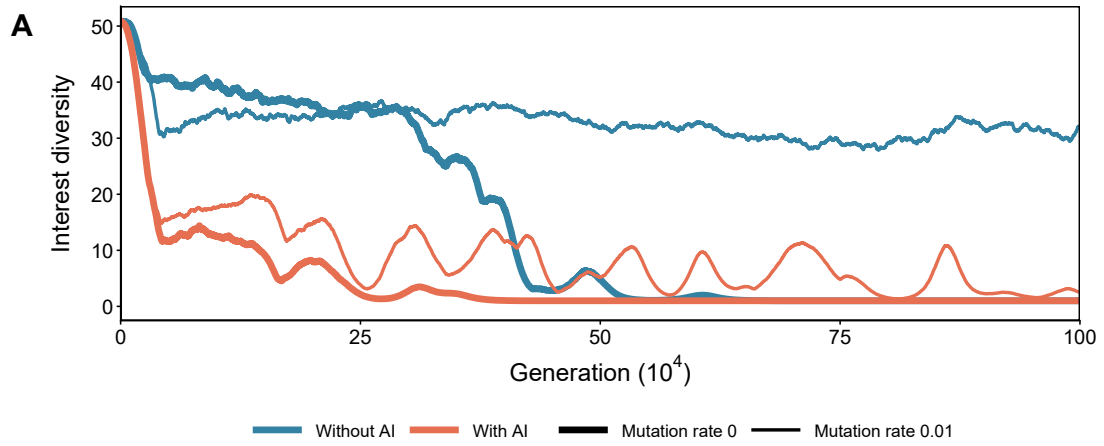


Figure S4: **Feedback structure becomes more prone to interest diversity loss with AI** Trajectories of interest diversity with and without AI under the feedback structure (A). Heatmaps of interest diversity across varying levels of mutation rate (B, C) and the standard deviation of the initial belief distribution (D, E), with and without AI. Interest diversity declines toward zero even without AI over long timescales when there is no mutation and the initial belief distribution is narrow. However, with AI, interest diversity declines more strongly, even under moderate mutation rates and wider initial belief distributions. Interest diversity in the heatmaps is averaged over generations 190,000–200,000. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$ for (D, E), $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$ for (A, B, C), and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.4$ and $\tau = 0.3$, chatbot AI is used.

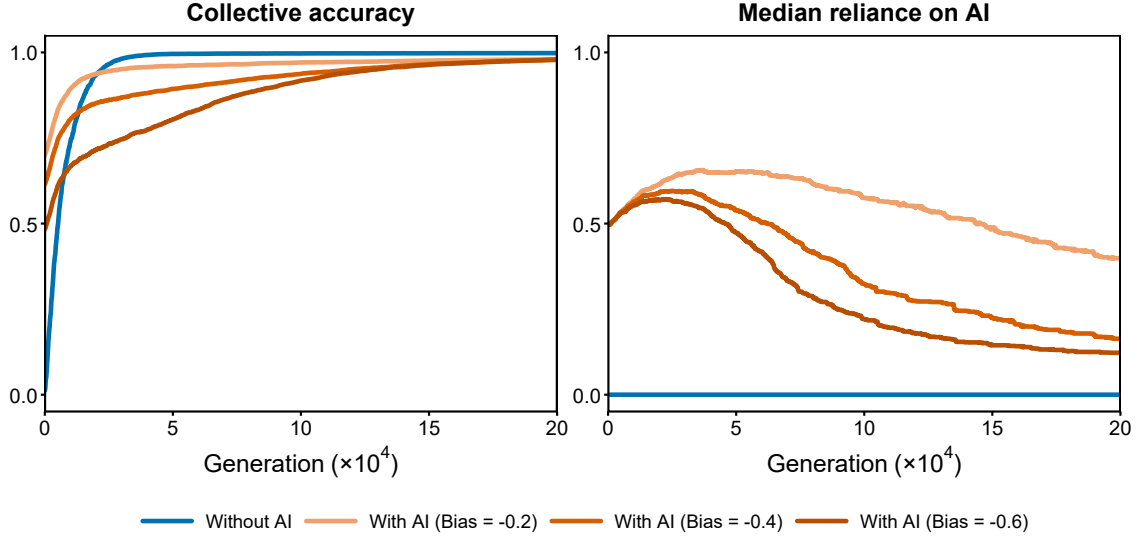


Figure S5: **Niche-expert structure may reduce reliance on AI, delaying the emergence of CI.** Trajectories of collective accuracy and median reliance on AI when the niche-expert structure causes reliance on AI to decline over time. Red lines represent trajectories with AI across varying levels of AI bias (-0.2, -0.4, and -0.6). Green lines represent trajectories without AI; reliance on AI is set to zero in the absence of AI. This reduction in reliance on AI occurs when the α_0 and AI bias have opposite signs, and is associated with the eventual emergence of CI, although CI emerges substantially later than in the condition without AI. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $\tau = 0.3$, chatbot AI is used.

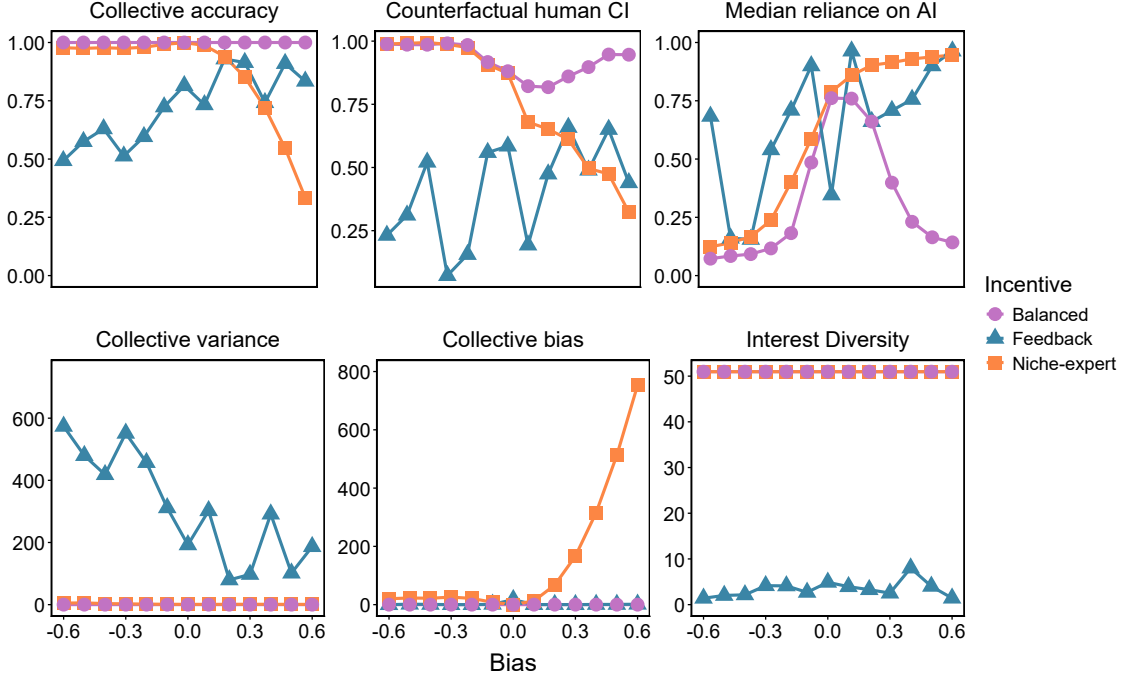


Figure S6: **Comparison of evaluation metrics between incentive structures across various levels of AI bias.** Evaluation metrics under feedback, niche-expert and balanced structures across varying levels of AI bias. Larger absolute AI bias indicates a less accurate AI model. Across the range of AI bias considered, AI accuracy varies from approximately 0.24 to nearly 1 when the AI predicts the target independently. All the evaluation metrics are averaged over generations 190,000–200,000. The balanced structure maintains near-perfect collective accuracy across bias levels and higher counterfactual human CI than the other incentive structures. These patterns are driven by well-calibrated reliance on AI, which increases as AI becomes more accurate, and by well-preserved interest diversity, which together lead to low collective variance and collective bias. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $\tau = 0.3$, chatbot AI is used.

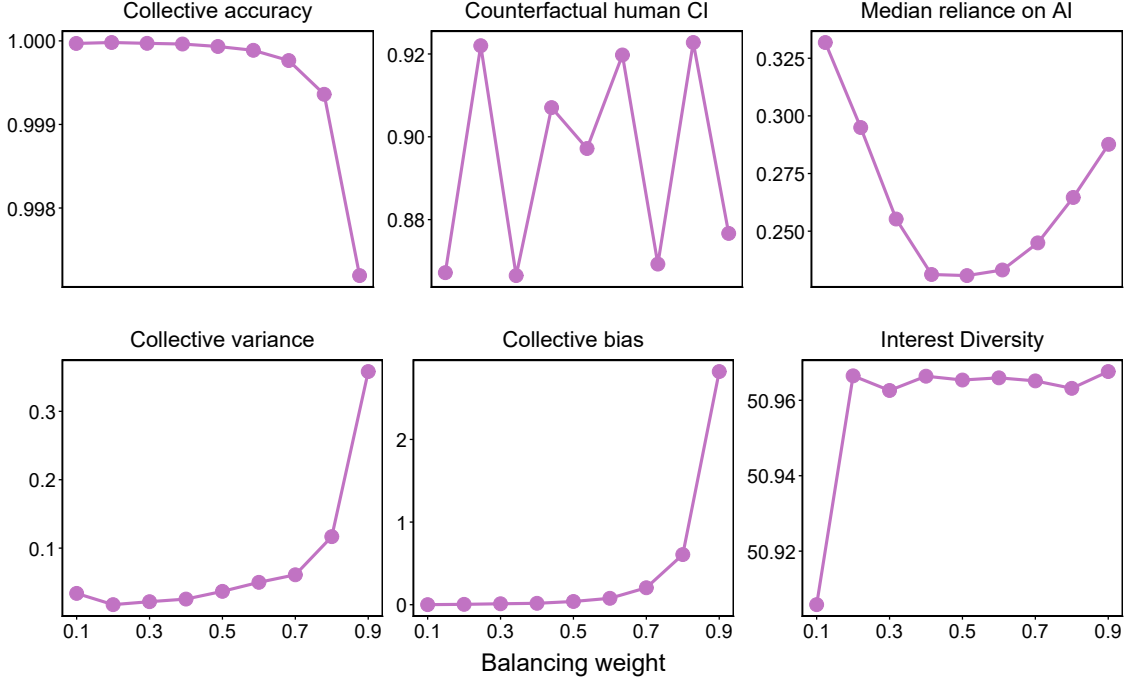


Figure S7: **Balanced structure is robust across varying balancing weights.** Evaluation metrics across varying levels of the balancing weight (0.1 – 0.9). The balancing weight represents the weight assigned to the niche-expert structure when constructing the balanced structure by combining the niche-expert and feedback structures. A balancing weight of 0.5 represents equal weighting of the niche-expert and feedback structures, whereas 0 corresponds to the feedback structure and 1 corresponds to the niche-expert structure. Although the evaluation metrics vary as the balancing weight changes from 0.1 to 0.9, the absolute magnitude of these deviations remains limited. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.4$, $\tau = 0.3$, chatbot AI is used.

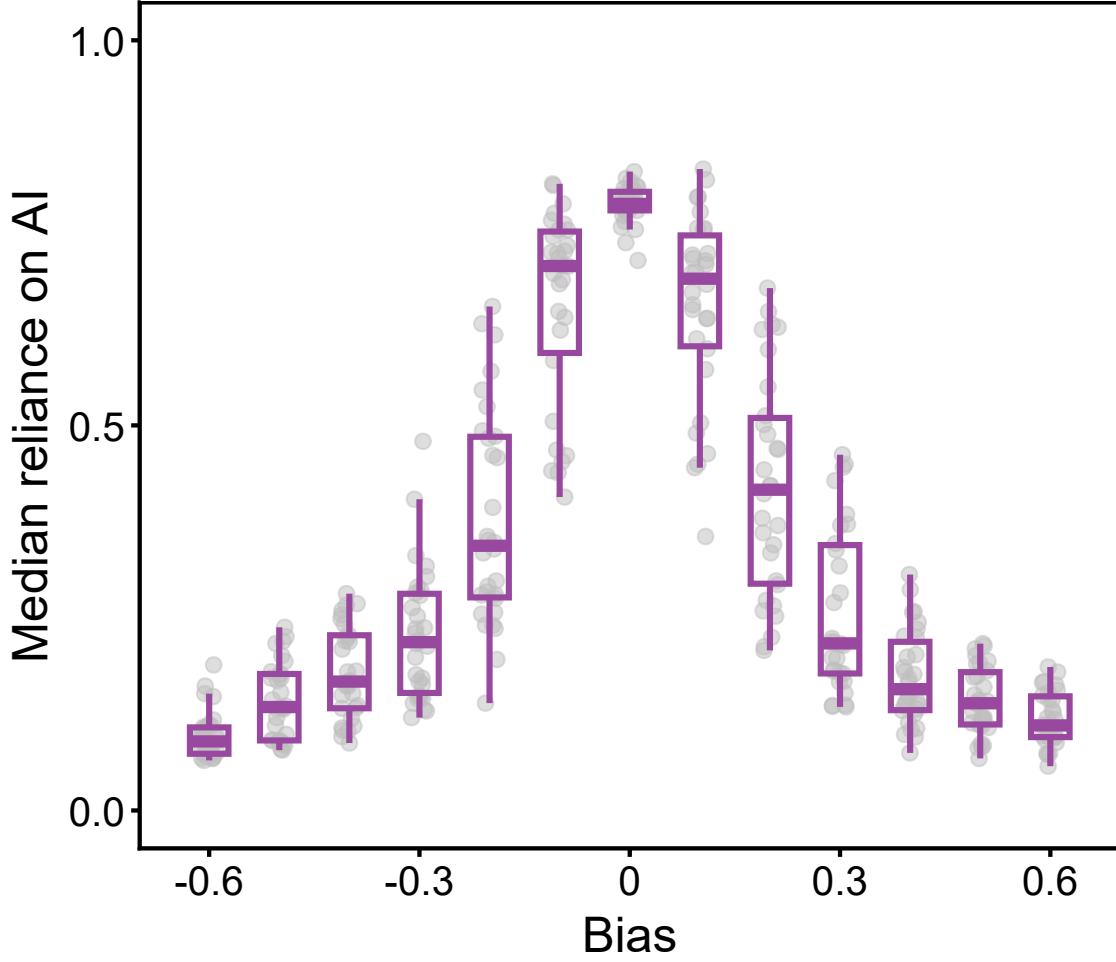


Figure S8: **Balanced structure shows well-calibrated reliance on AI.** Median reliance on AI across varying levels of AI bias. Thirty randomly initialized simulations were performed for each bias level. Each dot represents a single simulation, and the box plots show the median and interquartile range (IQR) for each bias level. The balanced structure shows well-calibrated reliance on AI, with reliance increasing as AI bias decreases. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $\tau = 0.3$, chatbot AI is used.

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